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A crude oil price ensemble forecasting system based on outlier correction and adaptive error-compensation strategy

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ABSTRACT

Crude oil price forecasting provides reliable information for energy policy formulation, corporate strategic planning, and investment decision-making. Its accuracy is often hampered by data outliers and inherent prediction errors, but many existing studies may overlook the significance of outlier preprocessing and error correction and often lack sufficient transparency. To address this issue, we propose a crude oil price ensemble forecasting system based on outlier correction and an adaptive error-compensation strategy. In this system, to identify and correct outliers in the original data, a data-preprocessing module is developed to eliminate negative impacts on model training and forecasting performance. In addition, an effective ensemble modelling module is proposed to combine the results of the diverse crude oil forecasting sub-models provided by the designed multiple-model forecasting module, which demonstrably outperforms other advanced ensemble methods and can perform better than any individual sub-model by fully harnessing their individual strengths. Another key advancement of this study is the proposal of an adaptive error-compensation strategy, which innovatively addresses not only how to correct errors in ensemble results but also whether such corrections are necessary. Moreover, the system incorporates an interpretable analysis module, yielding crucial insights and offering practical implications for decision-makers by enhancing the transparency of the ensemble system. Empirical studies show that the developed system has an average root mean squared error of 3.0810, representing an improvement of up to 11.53 % over the best sub-model. The system consistently outperforms the benchmark models and can be a promising solution for crude oil price forecasting.

Nomenclature

Data-preprocessing module, Stage I and Stage II of the framework			
$P_0(t)$	original crude oil price data at time t	$P_C(t)$	crude oil price data at time t after outlier detection and correction
$P_0^{med}(t)$	local median for the window around $P_0(t)$ in Hampel filter	$P_0^{MAD}(t)$	local median absolute deviation for the window around $P_0(t)$
$2i+1$	length of window	ξ	threshold of Hampel filter
$P_{\min}$, $P_{\max}$	minimum and maximum values of the data	$P(t)$	the value after data preprocessing
$P_{0-Train}$, P_{0-Val}, P_{0-Test}	original training, validation, and test datasets	P_{Train}, P_{Val}, P_{Test}	preprocessed training, validation, and test datasets
Stage III of the framework			

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(*continued*)

$P_{Train}(t)$	the value of t-th sample on training dataset	$\widehat{P}_{Val}$, $\widehat{P}_{Test}$	predictions from sub-models on validation and test sets
Multilayer perceptron			
$h^{(l-1)}$	input of l-th hidden layer	$h^{(l)}$	output of l-th hidden layer
$z^{(l)}$	pre-activation output of the l-th layer.	$S^{(l)}$	weight matrix of the l-th layer
$d^{(l)}$	bias vector of the l-th layer	$\sigma(\cdot)$	activation function
m_l	number of neurons in the l-th layer	m_{l-1}	number of neurons in the preceding layer
Extreme learning machine			
x_i	the i-th input	y_i	the i-th output
L	number of neurons in the hidden layer	ω_j	weights of the input layers

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(*continued*)

β_j	weights of output layers,	b_j	bias of the hidden layer
$g(\cdot)$	activation function		
Convolutional neural network			
$\mathbf{X}$	vector sequence of time series data	x_t	observed value at time t
T	total length of the sequence	$\mathbf{w}$	a convolutional kernel
k	length of the kernel	b	bias term of convolution
$\mathbf{y}$	feature map generated by convolution kernel	y_t	the t-th element of the convolution output
p	size of pooling window	z_t	the t-th element in the output of the pooling layer
Long short-term memory			
x_t	input at the current time step	h_{t-1}	hidden layer's state from the previous time step
W	weight matrix	b	bias parameter
$\widetilde{C}_t$	prospective state at the current time step	C_t	revised current cell state
f_t, i_t, o_t	outputs of the forget, input, and output gates at time t	$\sigma(\cdot)$	activation function
$\tanh(\cdot)$	hyperbolic tangent function		
Gated recurrent unit			
x_t	input at the current time step	h_{t-1}	hidden state from the preceding time step
W	weight matrix	b	bias parameter
z_t	proportion of the previous hidden state	r_t	the extent to which the previous hidden state is forgotten
$\widetilde{h}_t$	candidate hidden state for the current time step	h_t	ultimate hidden state at the current time step
Ensemble modeling module and Stage IV of the framework			
P	dependent variable	$\widehat{P}^1$	predictions from the first sub-model
λ_1	regression coefficients of the independent variables	λ_0	intercept of the ensemble model
ε	error term of the ensemble model	q	number of samples
$\mathbf{P}$	dependent variable matrix	$\widehat{\mathbf{P}}$	independent variable matrix
$\boldsymbol{\lambda}$	coefficients matrix	$\widehat{\mathrm{P}}_{Val}^{ELM}(n)$, $\widehat{\mathrm{P}}_{Test}^{ELM}(n)$	predictions of n-th sample from Extreme Learning Machine (ELM) on validation and test sets
λ_{ELM}	coefficient of ELM in the ensemble model	$\widehat{\mathrm{P}}_{Val}^{En}, \widehat{\mathrm{P}}_{Test}^{En}$	ensemble predictions on validation and test sets
Adaptive error correction module and Stage V of the framework			
$\widehat{P}(t)$	predicted value of at data point t	n	number of data points
$e(t)$	error at point t	$\bar{\lambda}_0$	intercept of the error correction model
ε	error term of the error correction model	$\widehat{\mathrm{P}}^C(t)$	corrected predicted values at point t
$e_{Val}^{En}, e_{Test}^{En}$	prediction error on validation and test sets	$e_{Val}^{En}(n)$, $e_{Test}^{En}(n)$	prediction error of n-th sample from ensemble on validation and test sets
e_{Test}^{C}	corrected error on test set	$e_{Test}^{C}(m)$	corrected error of m-th sample on test set
$\mathbf{D}$	matrix of the ensemble correction model	$\widehat{\mathrm{P}}_{Test}^{C}$	corrected predictions on test sets
Interpretable analysis module			
$\mho$	a model	N	index collection of all features
n	number of features	Ω	Shapley value
$\Omega_i(\mho)$	Shapley value for feature i	S	a subset of feature set N, excluding feature i.
$\mho(\cdot)$	predicted value of the model when using certain features		
Evaluation metrics			
y_i	true value of the i-th sample	$\widehat{y}_i$	predicted value of the i-th sample
$\bar{y}$	average of the true values of all samples	n	total number of samples

1. Introduction

1.1. Background

As one of the most critical resources in the global economy, crude oil serves as a cornerstone of industrial production and plays a pivotal role in international financial markets (Du et al., 2023). Fluctuations in oil prices reverberate beyond the economies of oil-producing and oil-consuming nations, triggering volatility across global markets (Jha et al., 2024). Recent events, such as the Russia–Ukraine conflict and the COVID-19 pandemic, have intensified this volatility (C. Xu et al., 2024; Zhang et al., 2024). Furthermore, traditional and renewable energy sources increasingly compete for market dominance as the shift toward renewables accelerates (Guo et al., 2021; Magazzino and Giolli, 2024), further intensifying the uncertainty in crude oil prices (Si Mohammed and Mellit, 2023). In particular, in 2023, global economic growth decelerated further, geopolitical situations became more severe and intricate, and global oil supply and demand, as well as trade patterns, underwent profound transformations. Against this complex backdrop, forecasting crude oil futures, which is essential for making informed energy investment decisions, effective risk management, and sound policymaking, has become increasingly challenging (Yang et al., 2024).

1.2. Literature review

Numerous scholars have investigated the prediction of crude oil prices, and their research has generally been classified into four major approaches: traditional econometric models, individual machine learning models, hybrid models, and ensemble models.

Commonly used econometric models include generalized autoregressive conditional heteroskedasticity, autoregressive integrated moving average, and Bayesian vector autoregressive models. Marchese et al. (2020) investigated several multivariate generalized autoregressive conditional heteroskedasticity models to predict the volatility and correlation between crude oil and its refined product. In the context of forecasting oil and other energy prices in India, Alam et al. (2023) utilized the autoregressive integrated moving average model and reported that it outperforms models such as the simple exponential smoothing model and the K-nearest neighbors model. Zhang et al. (2024) assessed the efficacy of various Bayesian vector autoregressive models for crude oil price forecasting across different time horizons and concluded that Bayesian vector autoregressive models deliver higher accuracy and robustness under complex economic conditions.

However, econometric methods rely on assumptions of linearity and stationarity (Zhang and Niu, 2024). Influenced by major events, crude oil price data are mostly nonstationary and nonlinear (Fang et al., 2023; Huang et al., 2023), and futures market volatility exhibits high-frequency characteristics (Chen et al., 2024). As the complexity of the global economic and political backdrop intensifies (Yu et al., 2024), linear patterns diminish, and nonlinear fluctuations become increasingly prominent. Consequently, many researchers have shifted their focus to machine learning models capable of effectively capturing the nonlinear characteristics inherent in data (Shen et al., 2025; Wu and Xie, 2025). Cen and Wang (2019) proposed a crude oil price forecasting method that utilizes long short-term memory (LSTM) networks, highlighting the superior ability of the model to capture complex fluctuation patterns inherent in time series data. In a related study, Zhao et al. (2023) employed LSTM for the early detection of exchange rate risks, focusing on the impacts of structural shocks in the international oil markets. Jin and Xu (2024) proposed a crude oil price prediction model based on nonlinear autoregressive neural networks for daily West Texas Intermediate (WTI) and Brent crude oil price prediction.

Although machine learning models have shown excellent performance, the improvement offered by a single model remains limited, and they are prone to pitfalls such as local optima and overfitting (Deng et al., 2023). Consequently, researchers have attempted to blend different methods for crude oil forecasting, called hybrid models, which have further enhanced the forecasting accuracy. For example, Wu et al. (2019) developed a hybrid crude oil price forecasting model by combining improved ensemble empirical mode decomposition with LSTM networks. Simsek et al. (2024) proposed a novel model that combines LSTM and extreme gradient boosting (XGBoost) algorithms to forecast crude oil prices. Shen et al. (2024) proposed a hybrid prediction model within a quantile regression framework, utilizing ensemble empirical mode decomposition to combine convolutional neural network (CNN) and bidirectional LSTM for the probabilistic forecasting of crude oil prices. Hybrid approaches have also been extended to multiperiod data. Jahandoost et al. (2024) developed a hybrid CNN-LSTM model that processes daily, weekly, and monthly data.

Because hybrid models rely on a single foundational model, their predictive capabilities, although enhanced by integrating various techniques, are ultimately constrained by the performance of the underlying model. This limitation hampers broader application and adoption. To address these challenges, researchers have developed ensemble models to overcome the shortcomings of hybrid models. Ensemble models combine diverse predictive models to fully leverage their strengths and overcome the limitations of each model, thereby improving overall prediction accuracy (Dalar and Egrioglu, 2025). Common ensemble methods are generally divided into two categories: weight-based methods and stacking-based methods (Chen and Liu, 2021). In weight-based models, a coefficient is typically assigned to each base model via simple averaging or optimization algorithms. For example, Safari and Davallou (2018) proposed a weight-based ensemble model for crude oil price forecasting that effectively enhances the prediction accuracy by combining the predictions of three individual models using both constant and time-varying weights. Hao et al. (2023) introduced a dynamic ensemble model based on weight allocation, incorporating a sliding window technique and a metabolic mechanism to dynamically adjust the weights of each model. In a separate study, Hao et al. (2022) proposed an ensemble strategy using the nondominated sorting genetic algorithm II intelligent optimization algorithm for weight generation. Similarly, Yang et al. (2023) employed four optimization algorithms to integrate sub-models for crude oil price forecasting. By contrast, stacking-based models utilize the predictions of one or more individual models as input features that are seamlessly integrated with another predictive model (Lin et al., 2025). For example, Qin et al. (2023) presented a tree-based stacking ensemble learning approach that integrates multiple tree-based models and leverages a stacking strategy to effectively combine them. Dai et al. (2024) and Xu et al. (2024) proposed ensemble models that utilize LSTM and random forest for stacking, respectively, designed for sea surface temperature and repose angle forecasting of railway ballasts. Xu et al. (2025) utilized LSTM as a nonlinear meta-learner to combine forecasts from various models and predict crude oil prices. The choice of the ensemble method is a critical step in ensemble forecasting. In general, equal-weight averaging simplifies the determination of weights, and its effectiveness may be limited. However, using artificial intelligence optimization algorithms to determine weights or employing stacking-based models may be more reasonable and effective than using an equal-weight averaging method. Nevertheless, whether an artificial intelligence optimization algorithm or an artificial neural network model is used, parameter tuning is a complex and challenging task that directly affects the ensemble prediction performance. Consequently, this study establishes a new ensemble method that overcomes the limitations of existing ensemble methods and improves crude oil price forecasting performance.

As shown in Table 1 and as mentioned above, various predictive models have advantages and disadvantages. Hence, ensemble forecasting models have emerged as promising approaches for crude oil price forecasting. In the data-preprocessing stage, existing research on crude oil price forecasting primarily involves data decomposition techniques and usually ignores the importance of outlier detection and correction. For example, Dong et al. (2024) used variational mode decomposition (VMD) to decompose an original time series into multiple intrinsic mode functions. More advanced decomposition strategies also exist, such as the two-layer multivariate variational mode decomposition (MVMD) proposed by Zhao et al. (2024), to decompose crude oil prices. However, as mentioned previously, crude oil prices are influenced by multiple factors and exhibit greater volatility in the current complex global economic and political context. Uncertain events may lead to abrupt fluctuations in market dynamics, resulting in outliers in the oil price series, which can directly affect model training and prediction performance. Without the proper detection and correction of these outliers, poor forecasting results may occur. Therefore, it is necessary to detect and correct outliers. In other areas of prediction, scholars have explored outlier detection and correction methods. However, few studies have explored this topic in the field of crude oil

Table 1
Overview of reviewed models.

Category	Authors	Advantage	Disadvantage	Whether to consider	
				Outlier	Error
Econometric	Marchese et al. (2020)	The econometric models are easy to train and implement.	Developed under the assumption of linearity; lacks the capability to capture nonlinear patterns in the data.	No	No
	Alam et al. (2023)			No	No
	Zhang et al. (2024)			No	No
Machine learning	Cen and Wang (2019)	Effectively identify and extract nonlinear features, leading to improved forecasting accuracy.	Contain numerous parameters, making it prone to getting stuck in local optima or overfitting.	No	No
	Zhao et al. (2023)			No	No
	Jin and Xu (2024)			No	No
Hybrid	Wu et al. (2019)	Incorporate the strengths of various methods to greatly enhance the forecasting performance of hybrid models.	The forecasting performance is heavily reliant on the individual prediction model.	No	No
	Simsek et al. (2024)			No	No
	Shen et al. (2024)			No	No
	Jahandoost et al. (2024)			No	No
Ensemble	Safari and Davallou (2018)	Incorporate multiple approaches to capture all relevant features of the original data, resulting in improved forecasting outcomes.	Establishing an effective and low complexity ensemble approach is challenging.	No	No
	Hao et al. (2023)			No	No
	Hao et al. (2022)			No	No
	Yang et al. (2023)			No	Yes
	Qin et al. (2023)			No	No
	Dai et al. (2024)			No	No
	Xu et al. (2024)			No	No
	Xu et al. (2025)			No	No
	Our work			Yes	Yes

price prediction. Ayiah-Mensah et al. (2025) employed the interquartile range (IQR) method to detect and adjust outliers in monthly rainfall data to improve the accuracy of seasonal rainfall forecasting. Shabbir et al. (2024) effectively integrated the Hampel filter (HF) into a hybrid approach for streamflow prediction to detect and correct outliers, highlighting its effectiveness in accurately capturing data trends and enhancing subsequent model prediction capabilities. HF is used to detect and handle outliers in time series data. It is based on statistical principles and has the advantages of robustness, strong adaptability, simple and intuitive parameters, data trend preservation, and local processing. Motivated by its successful application and advantages in handling outliers in complex time series data, in this study, a data-preprocessing module based on the HF is designed to detect and correct the outliers in the original crude oil price and to enhance the forecasting performance from the perspective of improving the modeling data quality.

In the field of crude oil price forecasting, extant studies have predominantly neglected the significant impact of model corrections on predictive performance. Although these forecasting models can enhance the accuracy to a certain extent, prediction errors remain inevitable owing to the influence of various factors. Nevertheless, by exploring and analyzing this error information, it is possible to further improve the predictive accuracy. In other forecasting fields, scholars have employed various error-correction strategies to more effectively extract valuable information from historical error sequences and reduce the impact of systematic errors on predictive accuracy. For example, Du et al. (2019) developed a container throughput forecasting model that employs an extreme learning machine (ELM) optimized via the butterfly optimization algorithm (to model errors via error sequences). Similarly, Du et al. (2022) proposed a residential electricity consumption forecasting model based on seasonal factors and a gray model with fractional-order accumulation optimized by the sine-cosine algorithm to model error sequences. However, after conducting experiments on crude oil price forecasting, we found that not all the error corrections were effective. This raises the important question of whether error correction is necessary for a specific prediction system and, if so, how to achieve effective error correction. To address this question, this study proposes an adaptive error-compensation strategy and developed an adaptive error-correction module that enables dynamic error correction to enhance the final forecasting performance.

1.3. Main work and contributions

On the basis of the abovementioned analysis and discussion, to better capture crude oil price changes and further improve predictive accuracy, this study proposes a new crude oil price ensemble forecasting system based on outlier correction and an adaptive error-compensation strategy. It comprises five modules: data preprocessing, multiple-model forecasting, ensemble modeling, adaptive error correction, and interpretable analysis. For convenience, the developed system is abbreviated as DMEAI, where D, M, E, A, and I represent the five aforementioned modules. Specifically, a data-preprocessing module was designed on the basis of the HF to identify and correct outliers in the original crude oil price time series, thereby reducing their disruptive impact on model training and forecasting performance. To provide diverse crude oil forecasting sub-models, a multiple-model forecasting module was proposed by introducing five different models: typical neural networks, multilayer perceptron (MLP), novel fast learning algorithms (ELM), and deep learning networks (CNN, LSTM, and gated recurrent unit (GRU)). In addition, unlike most previous ensemble methods that employ an artificial intelligence optimization algorithm or an artificial neural network model, this study innovatively proposes a novel ensemble modeling module to fully utilize the strengths of each sub-model and obtain initial ensemble forecasting results, which can achieve superior results to those of all sub-models. An adaptive error correction module is subsequently developed to determine whether error correction is conducted for the initial ensemble forecasting results and to realize adaptive error correction to improve the final forecasting performance if error correction is needed. Finally, an interpretable analysis module is designed on the basis of the Shapley additive explanation (SHAP), which focuses specifically on clarifying the mechanics of the ensemble stage, considering both pre- and post-error correction scenarios. At the same time, it provides decision-makers with clear explanatory opinions on the ensemble process and specific applications to enhance their understanding of how different modeling methods jointly inform predictions.

Compared with previous studies, the main innovations and contributions of this study are as follows.

(a) **A new crude oil price ensemble forecasting system was developed by introducing outlier correction and proposing an adaptive error-compensation strategy.** This fills the gap in crude oil price forecasting and uncovers the informational value of potential outliers and prediction errors. The ensemble forecasting system comprises five key modules: data preprocessing, multiple-model forecasting, ensemble modeling, adaptive error correction, and interpretable analysis. Experimental results on the Brent and WTI crude oil datasets demonstrate the superiority of the newly developed system over existing alternatives.

(b) **Outlier correction is introduced to enhance the forecasting performance from the perspective of the data.** Data preprocessing is crucial in crude oil price forecasting. However, previous studies have overlooked the significance of detecting and correcting data outliers, limiting further improvements in prediction accuracy. In this study, a data-preprocessing module is developed by incorporating outlier detection and correction techniques. Experimental results indicate that the data-preprocessing module can significantly mitigate the negative impact of outliers on model training, thereby enhancing the predictive performance.

(c) **A powerful ensemble modeling is designed to enhance the forecasting performance from the perspective of the model structure.** Prior studies on ensemble forecasting have primarily explored complex stacking models and optimization-based weighting strategies, which typically incur high computational and implementation burdens. By contrast, this study emphasizes diversity in the selection of sub-models, designs multiple-model forecasting based on diverse base learners, and proposes an effective yet less-complex ensemble method, which enables the system to effectively model various data characteristics and dynamics in crude oil price movements. The experimental results revealed that the ensemble modeling proposed in this study is superior to other advanced benchmarks and can improve the final forecasting performance.

(d) **A new adaptive error-compensation strategy is proposed to enhance the forecasting performance after the ensemble forecasting results are obtained.** Most existing studies have overlooked the importance of error correction in enhancing forecasting performance. A limited number of studies have explored this issue and demonstrated that applying error-correction techniques can effectively improve model accuracy, an approach that remains particularly rare in the context of crude oil price forecasting. Unlike the majority of previous studies, this study explicitly addresses the question of whether and how error correction should be implemented. To this end, this study proposes an adaptive error compensation strategy and develops it into an adaptive error correction module that consists of two steps: an error correction decision and an error compensation scheme.

(e) **Importance and interpretability analyses are performed to provide new evidence for crude oil price forecasting.** Unlike most previous studies that lack discussion on model importance and interpretability, this study analyzes the importance and interpretability of the ensemble system and explains the prediction behavior of the sub-models from the perspective of the whole sample. These analyses help identify and analyze the importance and contribution of the sub-models, enhance the interpretability of the ensemble system, and provide explanatory crude oil price forecasting analysis and decision support for decision-makers.

1.4. Structure of this paper

The remainder of this paper is organized as follows: Section 2 introduces the five key modules of the proposed crude oil price ensemble forecasting system and describes the overall framework, illustrating how these modules are integrated to predict crude oil prices. Section 3 presents the experimental design and the detailed results. Section 4 presents an in-depth discussion of the results. Finally, Section 5 concludes the paper.

2. Modules and framework of the proposed crude oil price ensemble forecasting system

This section details the proposed crude oil price ensemble forecasting system, including its five constituent modules and overall forecasting framework. Sections 2.1 to 2.5 introduce the core methodologies of these modules: the data preprocessing module for handling outliers and data normalization, the multiple-model forecasting module for generating diverse predictions, the ensemble modeling module for optimally combining the strengths of these sub-model, the adaptive error correction module for intelligently assessing the necessity of error correction and applying refinements to the ensemble results, and the interpretable analysis module for quantifying sub-model contributions to enhance the system's transparency. Finally, Section 2.6 presents the ensemble forecasting framework, detailing the workflow for applying the proposed system to crude oil forecasting.

2.1. Data preprocessing module

Effective data preprocessing is crucial for enhancing the quality of input data and subsequently improving the performance and robustness of forecasting models, especially for volatile time series, such as crude oil prices. The data preprocessing module in the proposed system comprises two primary stages: outlier detection and correction, and data normalization.

Stage I: Outlier detection and correction

To reduce the impact of outliers on model training, an outlier detection and correction stage based on the HF was developed, which can significantly increase the predictive performance. The HF algorithm was proposed by Hampel (1974) as a method for detecting and removing outliers from an input signal via the local median and median absolute deviation (MAD) for robust detection.

For a given data sequence $P_0(1), P_0(2), \ldots, P_0(n)$, we define a window of length $2i+1$ that encompasses the current point along with its i preceding and succeeding points. The local median, $P_0^{med}(t)$, is the median of all the data points within this window and can be calculated according to

$$P_0^{med}(t) = median(P_0(t-i), P_0(t-i+1), \ldots, P_0(t), \ldots, P_0(t+i-1), P_0(t+i)) \tag{1}$$

The MAD is defined as

$$P_0^{MAD}(t) = median\left(\left|P_0(t-i) - P_0^{med}(t)\right|, \ldots, \left|P_0(t+i) - P_0^{med}(t)\right|\right) \tag{2}$$

To determine whether a data point is an outlier, a threshold ξ was introduced. Specifically, if the absolute deviation of a data point was greater than that of ξ and $P_0^{MAD}(t)$, it was considered an outlier and corrected. The threshold ξ can be determined according to specific data conditions. The calculation rules were as follows:

$$P_C(t) = \begin{cases} P_0^{med}(t), & \dfrac{\left|P_0(t) - P_0^{med}(t)\right|}{P_0^{MAD}(t)} > \xi \\ P_0(t), & \dfrac{\left|P_0(t) - P_0^{med}(t)\right|}{P_0^{MAD}(t)} \leq \xi \end{cases} \tag{3}$$

Stage II: Data normalization

Following outlier detection and correction, a data normalization stage based on minimum–max normalization was employed. The core formula is as follows:

$$P(t) = \frac{P_C(t) - P_{\min}}{P_{\max} - P_{\min}} \tag{4}$$

where $P_{\min}$ and $P_{\max}$ are the minimum and maximum values of the data sequence, respectively, and where $P(t)$ is the resulting normalized value.

2.2. Multiple-model forecasting module

To furnish a variety of sub-models for crude oil forecasting, a multiple-model forecasting module was established by introducing five diverse models (MLP, ELM, CNN, LSTM, and GRU) that capture the hidden nature within the time series, thus establishing a solid foundation for an effective ensemble. The first category encompasses MLP (Yang et al., 2025), which represents a typical type of neural network and acts as the foundational structure for deep learning models. It learns complex representations by mapping the input data through multiple layers of nonlinear transformations. The second category includes a novel shallow neural network (ELM), which is known for its fast training speed and high computational efficiency (Yang et al., 2024), making it suitable for the rapid processing of large-scale data. The third category encompasses deep learning networks, such as CNN, LSTM, and GRU, where CNN (LeCun et al., 1998; Simsek et al., 2024) can extract local patterns effectively through convolutional operations, making them particularly adept at capturing local features within sequential data. LSTM (Almaghrabi et al., 2024) and GRU (Liu et al., 2024), the variants of the recurrent neural network (RNN) with explicit memory functions, can effectively address long-term dependencies present in time series datasets (Bao et al., 2025; Baz et al., 2024), thereby ensuring accurate temporal feature extraction.

2.2.1. MLP

MLP is a foundational feedforward artificial neural network consisting of an input layer, one or more hidden layers, and an output layer. By employing nonlinear activation functions, MLP can capture complex nonlinear relationships and has demonstrated significant efficacy in various fields, including time series forecasting (Adegboye et al., 2024). For the l-th hidden layer, the input is derived from the output of the preceding layer, denoted $h^{(l-1)}$, with the resulting output being $h^{(l)}$. The yield of each layer is computed through a linear transformation, followed by a nonlinear activation function that utilizes the output from the antecedent layer. The linear transformation and nonlinear activation functions are expressed in Eqs. (4) and (5), respectively.

$$z^{(l)} = S^{(l)} \cdot h^{(l-1)} + d^{(l)} \tag{5}$$

$$h^{(l)} = \sigma\left(z^{(l)}\right) \tag{6}$$

Here, $S^{(l)}$ represents the weight matrix of the l-th layer with dimensions m_l by m_{l-1}. m_l denotes the number of neurons in the l-th layer, and m_{l-1} signifies the number of neurons in the preceding layer. $d^{(l)}$ is the bias vector of the l-th layer with dimensionality m_l. The activation function $\sigma(\cdot)$ encompasses ReLU or sigmoid functions.

2.2.2. ELM

ELM, a single hidden layer neural network proposed by Huang et al. (2006), is renowned for its rapid learning speed owing to the random initialization of connection weights between the input and hidden layer biases (Hao et al., 2024a), which significantly accelerates the training process.

The key formula is

$$y_i = \sum_{j=1}^{L} \beta_j g\left(\omega_j x_i + b_j\right), i = 1, 2, \ldots, n \tag{7}$$

where ω_j and β_j represent the weights of the input and output layers, respectively, b_j is the bias of the hidden layer, $g(\cdot)$ is the activation function, and L indicates the number of neurons in the hidden layer.

2.2.3. CNN

CNN is a deep learning model with different types. One-dimensional CNNs are primarily used for sequence data processing (Hao et al., 2024b). The CNN is mainly composed of the following layers: the input layer, convolutional layer, activation function, pooling layer, and fully connected layer.

Given time series data, it can be represented as a vector sequence $\mathbf{x} = [x_1, x_2, \ldots, x_T]$, where x_t is the observed value at time t and T is the total length of the sequence. The convolutional layer extracts features by convolving the input sequence with a convolutional kernel. A convolutional kernel is a weight matrix known as $\mathbf{w} = [w_1, w_2, \ldots, w_k]$, where k represents the length of the kernel. The mathematical expression for the convolution operation is as follows.

$$y_t = \sum_{i=1}^{k} w_i \cdot x_{t+i-1} + b \tag{8}$$

where b is the bias term.

In the time series, the convolution kernel slides to generate a new feature map called the $\mathbf{y} = [y_1, y_2, \ldots, y_{T-k+1}]$. The activation function subsequently maps the linear combination output from the convolutional layer onto the nonlinear space. In the pooling layer, max pooling is commonly used to preserve the most prominent features of time series data. The maximum pooling formula is expressed by Eq. (9).

$$z_t = \max\left\{y_t, y_{t+1}, \ldots, y_{t+p-1}\right\} \tag{9}$$

where p denotes the pooling window size. After convolution and pooling operations, the fully connected layer generates the final output.

2.2.4. LSTM

Traditional RNNs often encounter directional loops during the information transmission process, which can easily lead to gradient explosion and gradient disappearance (Sharbatian and Moattar, 2023). LSTM, a special RNN, solves these problems (Hu et al., 2025). The LSTM units employ input, forget, and output gates to selectively control the writing, retention, and reading of information in their memory cells, thereby enabling the network to learn long-range dependencies from sequential data. The core principles of forget, input, and output gates are as follows:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{10}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{11}$$

$$\widetilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{12}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \tag{13}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{14}$$

$$h_t = o_t \cdot \tanh(C_t) \tag{15}$$

The input at the current time step is denoted as x_t, while h_{t-1} represents the hidden layer's state from the previous time step. Matrix W constitutes the weight matrix, and b represents the bias parameter. $\widetilde{C}_t$ denotes the prospective state at the current time step, and C_t represents the revised current cell state. The terms f_t, i_t, and o_t correspond to the outputs of the forget, input, and output gates, respectively, at time t. $\sigma(\cdot)$ denotes a sigmoid activation function, whereas $\tanh(\cdot)$ signifies a hyperbolic tangent function.

2.2.5. GRU

GRU, similar to LSTM, is designed to address the challenges of vanishing and exploding gradients within a conventional RNN (Zhang et al., 2023). This simplifies the complexity by incorporating only two gates: update and reset. The essence of this formulation is as follows.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \tag{16}$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \tag{17}$$

$$\widetilde{h}_t = \tanh(W_h \cdot [r_t \cdot \mathrm{h}_{t-1}, x_t] + b_h) \tag{18}$$

$$h_t = z_t \cdot h_{t-1} + (1 - z_t) \cdot \widetilde{h}_t \tag{19}$$

Here, akin to the LSTM, x_t represents the input at the current time step, whereas h_{t-1} is the hidden state from the preceding time step. The matrices W and the vector b are the weight matrix and bias vector associated with the update gate, respectively. The scalar z_t governs the proportion of the previous hidden state h_{t-1} that is retained, and r_t determines the extent to which the previous hidden state h_{t-1} is forgotten. $\widetilde{h}_t$ denotes the candidate hidden state for the current time step, and h_t is the ultimate hidden state at the current time step.

2.3. Ensemble modeling module

To fully leverage the strengths of each model within the multiple-model forecasting module and integrate their capabilities in capturing diverse features, an ensemble modeling module is developed. This module not only enhances the accuracy and stability of predictions, surpassing the performance of any sub-model but also addresses the limitations of individual models when dealing with complex data, ultimately yielding more comprehensive and precise forecasting outcomes. Inspired by weight-based ensemble forecasting methods, if the predictions from the multiple-model forecasting module are treated as independent variables $\left(\widehat{P}^1, \widehat{P}^2, \widehat{P}^3, \widehat{P}^4, \widehat{P}^5\right)$ and the actual values as the dependent variable P, the problem of ensemble modeling can be transformed into a linear regression problem:

$$P = \lambda_0 + \lambda_1 \widehat{P}^1 + \lambda_2 \widehat{P}^2 + \ldots + \lambda_5 \widehat{P}^5 + \varepsilon \tag{20}$$

where λ_0 represents the intercept, $\lambda_1, \lambda_2, \ldots, \lambda_5$ represents the regression coefficients of the independent variables, and ε denotes the error term, signifying the portion of the model that remains unexplained. When the number of samples is q, Eq. (20) in vector form can be condensed to

$$\mathbf{P} = \widehat{\mathbf{P}}\boldsymbol{\lambda} + \varepsilon$$
$$\widehat{\mathbf{P}} = \begin{bmatrix} 1 & \widehat{P}^1(1) & \cdots & \widehat{P}^5(1) \\ 1 & \widehat{P}^1(2) & \cdots & \widehat{P}^5(2) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \widehat{P}^1(q) & \cdots & \widehat{P}^5(q) \end{bmatrix} \tag{21}$$
$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \vdots \\ \lambda_p \end{bmatrix}, \mathbf{P} = \begin{bmatrix} P(1) \\ P(2) \\ \vdots \\ P(q) \end{bmatrix}$$

The crux for solving Eqs. (20) and (21) are based on the ordinary least squares (OLS) method. The objective of OLS is to determine the regression coefficients $\boldsymbol{\lambda}$ such that the sum of the squared residuals between the predicted and observed values is minimized, thereby minimizing the loss function, as depicted in

$$\min \sum_{i=1}^{q} (P_i - \widehat{P}_i)^2 = \min \|\mathbf{P} - \widehat{\mathbf{P}}\boldsymbol{\lambda}\|^2 \tag{22}$$

where $\widehat{P}_i = \widehat{\mathbf{P}}_i^T\boldsymbol{\lambda}$ represents the predicted value and $\mathbf{P} = [P(1), P(2), \ldots, P(q)]^T$ denotes the actual value.

2.4. Adaptive error correction module

Prediction errors are inevitable in forecasting systems. To extract valuable information from historical error sequences more effectively and further enhance the predictive performance, we propose an adaptive error compensation strategy. For each dataset, the strategy first tests whether error correction is necessary and then performs a correction if needed. On the basis of this strategy, an adaptive error-correction module was developed. Specifically, the module is composed of two parts: (I) the error correction decision and (II) the error compensation scheme. The former assesses the necessity of error correction, whereas the latter implements it.

2.4.1. Part I: error correction decision

The objective of error correction is to increase the accuracy of the predictions. However, the efficacy of correction strategies may vary across datasets with different characteristics. Therefore, it is unwise to apply error-correction strategies without considering them. The adaptive error correction module proposed in this study uses cross-validation of the validation set to determine whether the error compensation scheme should be applied to the test set. The specific steps are as follows:

Step 1: For the validation set, k-fold cross-validation is applied to divide the data into K subsets. Initially, we do not implement the Error Compensation Scheme detailed in Section 2.4.2 and calculate the average mean squared error (AMSE), which is denoted as $AMSE_0$. We subsequently apply the error compensation scheme detailed in Section 2.4.2 and compute the AMSE, labeled $AMSE_1$. Assuming that n data points exist, the AMSE is calculated as follows:

$$AMSE = \frac{1}{k}\sum_{i=1}^{k}\left(\frac{1}{n}\sum_{t=1}^{n}(P(t) - \widehat{P}(t))^2\right) \tag{23}$$

Step 2: If $AMSE_1$ is less than $AMSE_0$, then the error compensation scheme is applied. Otherwise, they should not be applied.

2.4.2. Part II: error compensation scheme

Part I addresses the necessity for error correction. However, for datasets requiring correction, the development of an effective correction strategy is a critical challenge. The error compensation scheme proposed in this study is outlined as follows:

Step 1: For a set of data points $P_0(1), P_0(2), \ldots, P_0(n)$ with the corresponding output from the ensemble modeling module values $\widehat{P}(1), \widehat{P}(2), \ldots, \widehat{P}(n)$, the calculation of the error at point t is as follows:

$$e(t) = P_0(t) - \widehat{P}(t) \tag{24}$$

Step 2: Having obtained an error sequence $(e(1), e(2), \ldots, e(n))$ through Step 1, an error correction model is established, where the independent variables remain $\left(\widehat{P}^1, \widehat{P}^2, \widehat{P}^3, \widehat{P}^4, \widehat{P}^5\right)$ and the dependent variable is now the error term e. For each point, the specifics are as follows:

$$e = \lambda_0 + \lambda_1\widehat{P}^1 + \lambda_2\widehat{P}^2 + \ldots + \lambda_5\widehat{P}^5 + \varepsilon \tag{25}$$

Upon training the error correction model with these n samples, one can then input the explanatory variables $\left(\widehat{P}^1, \widehat{P}^2, \widehat{P}^3, \widehat{P}^4, \widehat{P}^5\right)$ for each sample into the trained model, thereby obtaining the corrected error sequence $\left(e_1^C, e_2^C, \ldots, e_n^C\right)$.

Step 3: The corrected predicted values are equal to the output of the ensemble modeling module plus the corrected error, which can be expressed as

$$\widehat{P}^C(t) = \widehat{P}(t) + e^C(t) \tag{26}$$

where $\widehat{P}^C(t)$ represents the corrected predicted values at point t.

2.5. Interpretable analysis module

To enhance the interpretability of the forecasting system, an interpretable analysis module is established on the basis of the SHAP method introduced by Lundberg and Lee (2017). This module quantifies the contribution of each sub-model to ensemble forecasting, clearly illustrating the role and significance of each sub-model in overall forecasting. This aids in identifying the features that exert a substantial influence on the final decision-making process and enhances the transparency of the model, facilitating the comprehension of the model's decision-making procedures. Originating from cooperative game theory, SHAP was initially devised to measure members' contributions to the overall coalition (Shapley, 2016). The SHAP method aims to explain the forecasting results of a model by calculating the contribution of each feature to the output (Yao, 2025). Similar to the contributions of each individual in a team project, SHAP values fairly distribute the impact of each member on the ultimate success. The SHAP method assigns a measure, known as the SHAP value, to each feature of the predicted value for each sample. By subsequently aggregating the SHAP values across multiple samples, the global behavior of the entire model can be analyzed to identify the key features and their overall contributions to the model. The larger the absolute value of a feature's SHAP value is, the greater its contribution to the model; conversely, the smaller the absolute value, the lower its contribution. A positive SHAP value indicates a positive contribution, zero indicates no contribution, and a negative value indicates a negative contribution.

Suppose that we possess a model denoted as $\mho$ alongside a set of features, $N = \{1, 2, \ldots, n\}$. The Shapley value Ω was used to ascertain the average marginal contribution of each feature to the model output. The Shapley value for feature i is defined as follows:

$$\Omega_i(\mho) = \sum_{S \subseteq N\setminus\{i\}} \frac{|S|! \cdot (|N| - |S| - 1)!}{|N|!}[\mho(S \cup \{i\}) - \mho(S)] \tag{27}$$

where S represents a subset of feature set N, excluding feature i. $\mho(S\cup\{i\})$ denotes the model's predictive value when the feature set $\mho(S\cup\{i\})$ is utilized, whereas $\mho(S)$ signifies the model's predictive value when only the feature set $\mho(S)$ is used. The size of feature set N is denoted by $|N|$, and the size of subset S is denoted by $|S|$.

2.6. Framework of the proposed crude oil price ensemble forecasting system

This section presents the overall forecasting framework based on the aforementioned modules. Stage I describes the data collection and reconstruction, and Stages II to VI detail the systematic workflow of the proposed system for crude oil price forecasting. Fig. 1 shows a flowchart illustrating the proposed forecasting system.

Stage I: Oil price data collection and reconstruction

To evaluate the performance of the developed forecasting system, weekly futures prices for WTI and Brent crude oil, as shown in Fig. 2, were utilized in this study. All datasets are sourced from https://www.investing.com, a prominent financial platform that provides information on futures, stocks, and other financial instruments. The date range for each dataset spans from January 1, 2016, to December 31, 2023. The data for January 3, 2016–May 8, 2022, were designated the training set $P_{0-Train}$;those for May 15, 2022–August 13, 2023, were designated the validation set P_{0-Val}; and those for August 20, 2023, to December 31, 2023, were designated the test set P_{0-Test}. Table 2 lists the statistical values for each experimental dataset. According to the two-sample Kolmogorov–Smirnov test statistical value and p value, the two datasets exhibit significant differences, which further reveals that the conclusions of the empirical studies are convincing.

Stage II: Original data preprocessing
Step I: Outlier detection and correction

As discussed above, crude oil prices may cause outliers for many reasons, such as uncertain events, which can directly affect model training and prediction performance. Therefore, at this stage, a data preprocessing module is proposed to realize outlier detection and correction, thereby mitigating the impact of outliers on model training and improving predictive performance. The original data were preprocessed with a window size of 5 and a threshold of 2.5. For value $P_0(t)$ at time t, the revised value $P_C(t)$ is as follows:

$$
\begin{aligned}
&P_C(t) = \begin{cases} P_0^{med}(t) & ,\left|P_0(t) - P_0^{med}(t)\right| < 2.5{\cdot}P_0^{MAD}(t) \\ P_0(t) & ,\left|P_0(t) - P_0^{med}(t)\right| \le 2.5{\cdot}P_0^{MAD}(t) \end{cases} \\
&P_0^{med}(t) = median(P_0(t-2), P_0(t-1), P_0(t), P_0(t+1), P_0(t+2)) \\
&P_0^{MAD}(t) = median\left(\left|P_0(t-2) - P_0^{med}(t)\right|, \dots, \left|P_0(t+2) - P_0^{med}(t)\right|\right)
\end{aligned}
\tag{28}
$$

Step II: Data normalization

For the data processed in the previous step, min–max normalization was performed before they were used for model training. The preprocessed data for training, validation, and testing were labeled P_{Train}, P_{Val}, and P_{Test}, respectively.

Stage III: Multiple-model pretraining and forecasting

To capitalize on the strengths of the different sub-models in handling various data, a multiple model forecasting module was developed at this stage, utilizing the WTI and Brent data processed in the previous stage for preliminary predictions. This stage lays the foundation for the subsequent ensemble stage. All the predictions are single-step predictions using a scrolling mechanism with a window size of five. For example, within the training set, to predict $P_{Train}(t)$, a time series $\{P_{Train}(t-5), P_{Train}(t-4), P_{Train}(t-3), P_{Train}(t-2), P_{Train}(t-1)\}$ is used for the forecast. The window is subsequently shifted forward by one time step, employing a time series $\{P_{Train}(t-4), P_{Train}(t-3), P_{Train}(t-2), P_{Train}(t-1), P_{Train}(t)\}$ to predict $P_{Train}(t+1)$. At this stage, by using the pretrained model and applying denormalization to its outputs, the prediction values $\widehat{P}_{Val}$ for the validation set and $\widehat{P}_{Test}$ for the test set can be obtained. Taking WTI as an example, the prediction results of five pretrained models for the validation set are denoted as $\widehat{P}_{Val}^{ELM}$, $\widehat{P}_{Val}^{MLP}$, $\widehat{P}_{Val}^{CNN}$, $\widehat{P}_{Val}^{LSTM}$ and $\widehat{P}_{Val}^{GRU}$, whereas those for the test set are termed $\widehat{P}_{Test}^{ELM}$, $\widehat{P}_{Test}^{MLP}$,

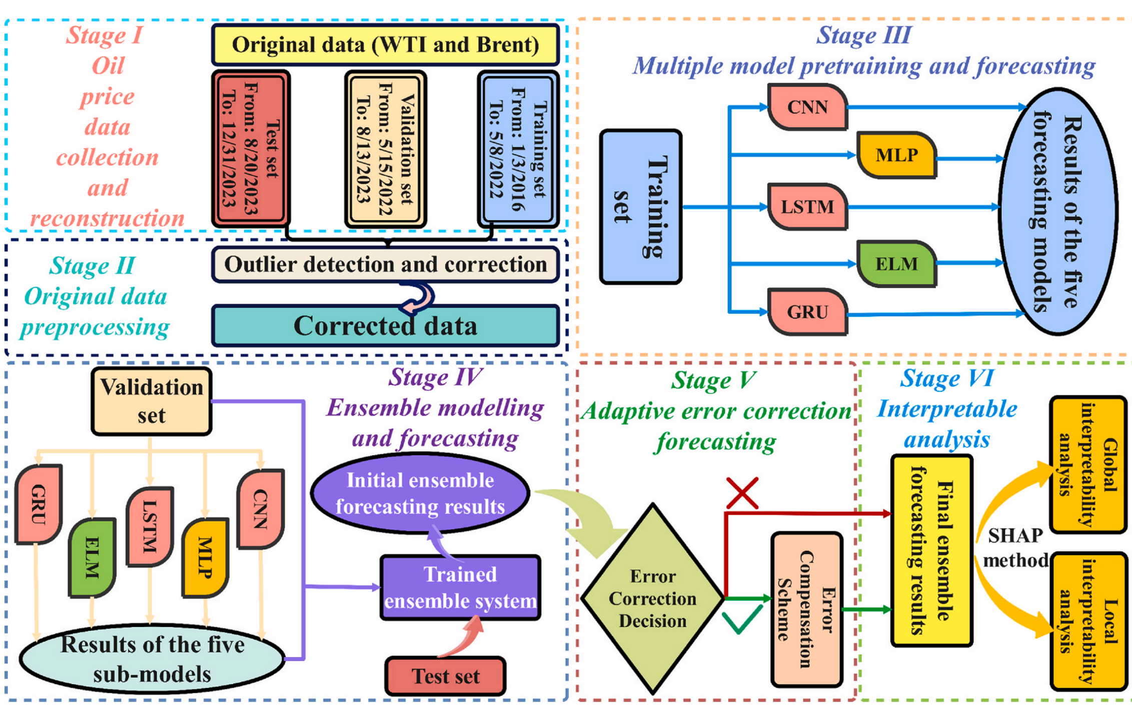

Fig. 1. Flowchart of the proposed forecasting system.

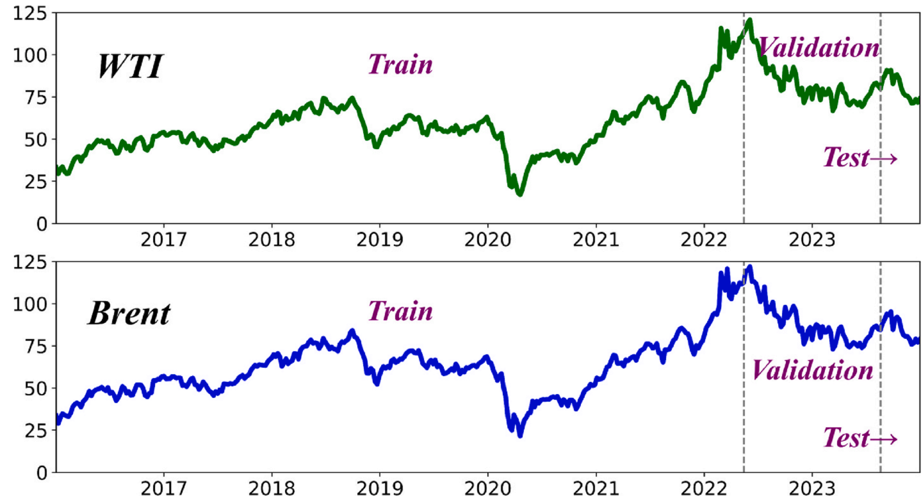

Fig. 2. Experimental data of WTI and Brent.

Table 2
Statistical characteristics of the studied data.

Data	Part	Number	Min.	Max.	Median	Ave.	Std.	Skewness	Kurtosis
WTI	All	418	16.9400	120.6700	59.2800	62.0971	18.7807	0.5388	0.3399
	Train	332	16.9400	115.6800	54.3450	56.6182	16.1752	0.8570	1.8801
	Validation	66	66.7400	120.6700	80.2050	84.0288	12.9576	1.2381	0.9460
	Test	20	71.2300	90.7900	80.1700	80.6720	7.0867	0.0943	−1.6136
Brent	All	418	21.4400	122.0100	65.2350	66.5356	19.2793	0.3920	0.0113
	Train	332	21.4400	120.6500	61.5900	60.9144	16.7247	0.6611	1.2014
	Validation	66	72.9700	122.0100	86.0300	89.2480	12.6082	0.9710	0.1964
	Test	20	75.8400	95.3100	84.5300	84.8975	6.4766	0.1735	−1.4965
Two-sample Kolmogorov−Smirnov test				**ks2stat**	**0.1388**	**P value**	**0.0006**		

Table 3
Key parameters and hyperparameter settings.

Model	Parameter/Hyperparameter	Values for WTI/Brent
MLP	hidden layers	2/2
	neurons per hidden layer	64/64
	activation function	sigmoid/sigmoid
	batch size	32/32
	epochs	400/400
	patience	15/15
ELM	neurons of hidden layer	100/100
	activation function	sigmoid/sigmoid
CNN	Conv1D filters	64/64
	kernel size	3/3
	Activation (convolutional layer)	relu/relu
	MaxPooling1D	2/2
	batch size	16/16
	epochs	500/500
	patience	15/15
LSTM	number of layers	2/2
	number of LSTM units	24/24
	batch size	32/32
	epochs	400/400
	patience	15/15
GRU	number of layers	1/1
	number of GRU units	64/64
	batch size	64/64
	epochs	400/400
	patience	15/15

$\widehat{\mathrm{P}}_{Test}^{CNN}$, $\widehat{\mathrm{P}}_{Test}^{LSTM}$ and $\widehat{\mathrm{P}}_{Test}^{GRU}$. Table 3 summarizes the key parameters and hyperparameter settings for the individual sub-models.

Stage IV: Ensemble modeling and forecasting

To fully harness the capabilities of each sub-model in processing diverse data, an ensemble modeling module is proposed at this stage, which not only enhances the accuracy and stability of predictions but also addresses the limitations of individual models when handling complex data, ultimately leading to more comprehensive and precise forecasting outcomes. Taking WTI as an example, the prediction values $\widehat{\mathrm{P}}_{Val}^{ELM}$, $\widehat{\mathrm{P}}_{Val}^{MLP}$, $\widehat{\mathrm{P}}_{Val}^{CNN}$, $\widehat{\mathrm{P}}_{Val}^{LSTM}$, and $\widehat{\mathrm{P}}_{Val}^{GRU}$ obtained from **Stage III** for the validation set were utilized in this stage to train the ensemble model. Specifically, the predictions obtained from the five sub-models were considered features, and the original crude oil price preprocessed by the data preprocessing module was considered the target for constructing a regression model, as follows:

$$\begin{bmatrix} \mathrm{P}_{Val}(1) \\ \mathrm{P}_{Val}(2) \\ \vdots \\ \mathrm{P}_{Val}(n) \end{bmatrix} = \begin{bmatrix} 1 & \widehat{\mathrm{P}}_{Val}^{ELM}(1) & \widehat{\mathrm{P}}_{Val}^{MLP}(1) & \widehat{\mathrm{P}}_{Val}^{CNN}(1) & \widehat{\mathrm{P}}_{Val}^{LSTM}(1) & \widehat{\mathrm{P}}_{Val}^{GRU}(1) \\ 1 & \widehat{\mathrm{P}}_{Val}^{ELM}(2) & \widehat{\mathrm{P}}_{Val}^{MLP}(2) & \widehat{\mathrm{P}}_{Val}^{CNN}(2) & \widehat{\mathrm{P}}_{Val}^{LSTM}(2) & \widehat{\mathrm{P}}_{Val}^{GRU}(2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \widehat{\mathrm{P}}_{Val}^{ELM}(n) & \widehat{\mathrm{P}}_{Val}^{MLP}(n) & \widehat{\mathrm{P}}_{Val}^{CNN}(n) & \widehat{\mathrm{P}}_{Val}^{LSTM}(n) & \widehat{\mathrm{P}}_{Val}^{GRU}(n) \end{bmatrix} \begin{bmatrix} \lambda_0 \\ \lambda_{ELM} \\ \vdots \\ \lambda_{GRU} \end{bmatrix} \tag{29}$$

where $\boldsymbol{\lambda} = [\lambda_0, \lambda_{ELM}, \ldots, \lambda_{GRU}]^T$ represents the intercept and coefficient of each sub-model in the ensemble model, with specific values determined through the application of OLS. Subsequently, on the basis of the well-fitted ensemble model, according to $\widehat{\mathrm{P}}_{Test}^{ELM}$, $\widehat{\mathrm{P}}_{Test}^{MLP}$, $\widehat{\mathrm{P}}_{Test}^{CNN}$, $\widehat{\mathrm{P}}_{Test}^{LSTM}$, and $\widehat{\mathrm{P}}_{Test}^{GRU}$ obtained in **Stage III**, the ensemble forecast result can be obtained and is referred to as $\widehat{\mathrm{P}}_{Test}^{En}$.

Stage V: Adaptive error correction forecasting

To better mine valuable information from historical error patterns and improve predictive accuracy, an adaptive error correction module is implemented at this stage. Initially, the error correction decision was designed to assess the necessity of error correction and determine whether error correction should be conducted for the initial ensemble forecasting results. Subsequently, for datasets that warrant error correction, an error compensation scheme is implemented to realize adaptive error correction and improve the final forecasting performance. The specific steps were as follows:

Step 1: Construct the error sequence on the validation set

The prediction error for the validation set's predicted values $\widehat{P}^{En}_{Val}$ is given by

$$e^{En}_{Val} = P_{0-Val} - \widehat{P}^{En}_{Val} \tag{30}$$

where P_{0-Val} is the original value obtained without data preprocessing. For n data points in the validation set, the error sequence can be constructed as $\left(e^{En}_{Val}(1), e^{En}_{Val}(2), \ldots, e^{En}_{Val}(n)\right)$.

Step 2: Establish the error compensation model on the validation set

The features of the error correction model consisted of forecasting values from the five sub-models on the validation set $\left(\widehat{P}^{ELM}_{Val}, \widehat{P}^{MLP}_{Val}, \widehat{P}^{CNN}_{Val}, \widehat{P}^{LSTM}_{Val}, \widehat{P}^{GRU}_{Val}\right)$, with the target being the prediction error from the ensemble modeling module e^{En}_{Val}. The mathematical formulas are as follows:

$$\begin{bmatrix} e^{En}_{Val}(1) \\ e^{En}_{Val}(2) \\ \vdots \\ e^{En}_{Val}(n) \end{bmatrix} = \begin{bmatrix} 1 & \widehat{P}^{ELM}_{Val}(1) & \widehat{P}^{MLP}_{Val}(1) & \widehat{P}^{CNN}_{Val}(1) & \widehat{P}^{LSTM}_{Val}(1) & \widehat{P}^{GRU}_{Val}(1) \\ 1 & \widehat{P}^{ELM}_{Val}(2) & \widehat{P}^{MLP}_{Val}(2) & \widehat{P}^{CNN}_{Val}(2) & \widehat{P}^{LSTM}_{Val}(2) & \widehat{P}^{GRU}_{Val}(2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \widehat{P}^{ELM}_{Val}(n) & \widehat{P}^{MLP}_{Val}(n) & \widehat{P}^{CNN}_{Val}(n) & \widehat{P}^{LSTM}_{Val}(n) & \widehat{P}^{GRU}_{Val}(n) \end{bmatrix} \begin{bmatrix} \lambda_0 \\ \lambda_{ELM} \\ \vdots \\ \lambda_{GRU} \end{bmatrix} \tag{31}$$

where the matrix $\mathbf{D} = \left[\lambda^{WTI}_0, \lambda^{WTI}_{ELM}, \ldots, \lambda^{WTI}_{GRU}\right]^T$ represents the intercept and the weight of each feature in the error correction model, and the specific values are determined through OLS.

Step 3: Error correction decision

If $AMSE_0$ is less than $AMSE_1$, no error correction is needed, and the ensemble forecasting results serve as the final forecast. If $AMSE_1$ is less than $AMSE_0$, we apply an error correction and proceed to the next step.

Step 4: Obtain the corrected prediction values

Applying the well-fitted error correction model to the test set with m data points yields the corrected error sequence, $\left(e^{C}_{Test}(1), e^{C}_{Test}(2), \ldots, e^{C}_{Test}(m)\right)$.

The corrected prediction value $\widehat{P}^{C}_{Test}$ can be obtained by

$$\widehat{P}^{C}_{Test} = \widehat{P}^{En}_{Test} + e^{C}_{Test} \tag{32}$$

Stage VI: Interpretable analysis

Once the final predictive values are computed, to gain a deeper understanding of the model, the interpretable analysis module is used to understand the contribution and significance of each sub-model to the prediction outcome. This includes both global interpretability analyses and local interpretability analyses. Global interpretability analysis focuses on the overall picture, measuring each sub-model's contribution holistically and ranking model importance on the basis of these contributions. Local interpretability analysis, on the other hand, focuses on specific samples, clarifying the decision-making process for each sample and, specifically, how each sub-model contributes to the predicted value. The ranking of sub-model importance may vary across samples.

3. Empirical studies

This section provides comprehensive empirical studies of the proposed forecasting system, including the experimental design, performance metrics, and analysis of the experimental results.

3.1. Experiment design

In the empirical studies, a series of comprehensive experiments were designed to systematically validate the proposed crude oil price ensemble forecasting system and to rigorously evaluate the contributions of its core components. The primary objective of this experimental design was to provide a structured, evidence-based assessment of the efficacy of each key module: data preprocessing (D), ensemble modeling (E) based on multiple model forecasting (M), and adaptive error correction (A). Specifically, five experiments were designed as follows: Experiment I aimed to assess how effectively the data preprocessing module improved the forecasting accuracy by detecting and correcting outliers. Experiment II focused on validating the effectiveness of the ensemble forecasting module and Experiment III focused on demonstrating the superiority of the developed system by comparing the proposed ensemble module with various advanced ensemble approaches, including a state-of-the-art model-based ensemble method. Experiment IV aimed to demonstrate the performance improvement achieved by the adaptive error correction module, which can verify the effectiveness of the proposed adaptive error correction module. Finally, in Experiment V, an ablation study was designed to quantify the performance degradation resulting from the removal or replacement of key modules, which further illustrates the essential contributions of each module and the superiority of the developed system.

3.2. Performance metrics

Six commonly used evaluation metrics are listed in Table 4: the root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), normalized mean squared error (NMSE), index of agreement (IA), and Theil's U1 statistic, which were selected to assess the predictive performance of all the models. The RMSE and MAE were used to measure the absolute size of the errors, whereas the MAPE and U1 focused on the relative percentage of errors, with U1 considering the relative change between the predicted and actual values. All these metrics are better when they are close to zero. The NMSE considers the

Table 4
Prediction performance evaluation metrics.

Metrics	Equations
RMSE	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}$
MAE	$MAE = \frac{1}{n}\sum_{i=1}^{n}\left\lvert y_i - \widehat{y}_i\right\rvert$
MAPE	$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \widehat{y}_i}{y_i}\right\rvert \times 100\%$
U1	$U1 = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}y_i^2} + \sqrt{\frac{1}{n}\sum_{i=1}^{n}\widehat{y}_i^2}}$
NMSE	$NMSE = \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2}$
IA	$IA = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(\left\lvert y_i - \overline{y}\right\rvert + \left\lvert\widehat{y}_i - \overline{y}\right\rvert\right)^2}$

variance of the true values and can compare the model performance by simply using the mean as the prediction; the NMSE is better when it is smaller. If the NMSE is greater than one, the model's performance is worse than when the mean is used as the prediction. The IA measures the consistency between the predicted and actual values, reflecting whether the model's predictions are consistent across different samples, with a value closer to one indicating better consistency.

3.3. Experiment I: effectiveness of the designed data preprocessing module

As described in the proposed forecasting system, the newly designed data preprocessing module can identify and correct outliers, which can enhance the predictive performance. To empirically validate its utility, a comparative experiment was conducted to assess the predictive performance of multiple models with and without data preprocessing. The detailed results of the different models are presented in Table 5, and Fig. 3. The detailed analysis is as follows.

(a) The experimental findings indicated that across all the datasets, the application of the data preprocessing module consistently yielded superior prediction results for each model. The performance evaluation metrics of the models with data preprocessing, including the RMSE, MAE, MAPE, U1, and NMSE, are all reduced. In addition, the degree of improvement varied across the models. Taking Brent as an example, the enhancements in the evaluation metrics were distributed as follows: the decrease in the RMSE was between 2.2972 % and 39.1788 %, the MAE reduction ranged from 2.1293 % to 43.3401 %, the decrease in the MAPE ranged from 2.9031 % to 45.9810 %, the U1 improvement was between 2.6936 % and 40.9677 %, and the NMSE reduction was between 4.5328 % and 63.0044 %. Concurrently, the IA was enhanced, with an increase ranging from 1.5405 % to 15.1308 % in Brent, reflecting an overall improvement in predictive performance compared with models without data preprocessing. Among these models, the MLP model demonstrated the most substantial improvement, with percentage enhancements for each evaluation criterion recorded for $I_{MLP}^{RMSE} = 39.1788\%$, $I_{MLP}^{MAE} = 43.3401\%$, $I_{MLP}^{MAPE} = 45.9810\%$, $I_{MLP}^{U1} = 40.9677\%$, $I_{MLP}^{NMSE} = 63.0044\%$, and $I_{MLP}^{IA} = 15.1308\%$.

(b) Without data preprocessing, ELM performed the best on WTI, with an RMSE of 3.7372, whereas MLP performed the worst, with an RMSE of 8.3895. For Brent, the ELM performed the best, with an RMSE of 3.7542, whereas the MLP performed the worst, with an RMSE of 5.1584. With data preprocessing, ELM performed the best on the WTI, achieving an RMSE of 3.4929, whereas MLP performed the worst, with an RMSE of 6.6252. However, for Brent, MLP outperformed the other models, achieving the best RMSE of 3.1374, whereas CNN exhibited the worst performance, with an RMSE of 4.9379. These results underscore the fact that no single model is universally superior across datasets. Furthermore, they highlight the importance of data preprocessing and the integration of various models to achieve enhanced predictive performance.

Remark: The data preprocessing module effectively identifies and corrects outliers in the original data, thereby comprehensively enhancing the predictive performance. Moreover, no single model consistently outperforms the others when dealing with diverse data, highlighting the limitations of individual models and underscoring the need for ensemble models.

3.4. Experiment II: effectiveness of the designed ensemble modeling module

By integrating the five sub-models, the unique capabilities of each model can be harnessed to address data with varying characteristics. The next step was to validate the superiority of the proposed ensemble system over the individual sub-models. **Experiment I** confirmed a significant improvement in the data preprocessing module. Therefore, to maintain control over the variables, all the models utilized data from a data preprocessing module for further testing. A comparison of the results is presented in Table 6 and Fig. 4.

On the basis of the results presented in Table 6, the proposed ensemble system consistently outperformed each individual sub-model across all evaluated performance metrics, highlighting its ability to effectively integrate the strengths of its components for more robust and reliable predictions. For WTI, the proposed ensemble system achieved RMSEs of 3.0900, 2.4903, and 3.0616 and U1 values of 0.0191, 0.1901, and 0.9409, respectively. Compared with the best-performing sub-model, these metrics represent improvements of 11.5348 %, 11.8072 %, 10.6624 %, 10.7477 %, 21.7373 %, and 1.0091 %, respectively. Compared with those of the worst sub-model, the enhancements were even more substantial, with improvements of 53.3599 %, 53.5712 %, 55.4696 %, 54.3062 %, 78.2494 %, and 65.6514 %, respectively. For

Table 5
Prediction results of multiple models with/without data preprocessing.

Dataset	Model	Model performance					
		RMSE	MAE	MAPE(%)	U1	NMSE	IA
WTI	MLP	8.3895	6.7698	8.9185	0.0535	1.4015	0.4594
	D-MLP	**6.6252**	**5.3637**	**6.8753**	**0.0418**	**0.8740**	**0.5680**
	ELM	3.7372	2.9446	3.5313	0.0229	0.2781	0.9222
	D-ELM	**3.4929**	**2.8237**	**3.4270**	**0.0214**	**0.2429**	**0.9315**
	CNN	5.1536	4.1562	5.0994	0.0318	0.5289	0.7942
	D-CNN	**4.9389**	**4.0499**	**4.9728**	**0.0305**	**0.4857**	**0.8133**
	LSTM	5.2270	4.5269	5.4642	0.0320	0.5440	0.7951
	D-LSTM	**4.9738**	**4.3808**	**5.3153**	**0.0305**	**0.4926**	**0.8063**
	GRU	4.4167	3.6729	4.5137	0.0273	0.3884	0.8520
	D-GRU	**4.1490**	**3.4545**	**4.2388**	**0.0256**	**0.3428**	**0.8730**
Brent	MLP	5.1584	4.2741	5.2028	0.0310	0.6344	0.8063
	D-MLP	**3.1374**	**2.4217**	**2.8105**	**0.0183**	**0.2347**	**0.9283**
	ELM	3.7542	2.7188	3.1022	0.0219	0.3360	0.9014
	D-ELM	**3.3782**	**2.5688**	**2.9629**	**0.0197**	**0.2721**	**0.9219**
	CNN	5.0540	4.1422	4.8362	0.0297	0.6089	0.7595
	D-CNN	**4.9379**	**4.0540**	**4.6958**	**0.0289**	**0.5813**	**0.7712**
	LSTM	4.9496	4.1138	4.7380	0.0289	0.5840	0.7714
	D-LSTM	**4.6268**	**3.9078**	**4.4847**	**0.0270**	**0.5103**	**0.8095**
	GRU	4.1438	3.5093	4.0756	0.0243	0.4094	0.8429
	D-GRU	**3.9178**	**3.1792**	**3.6452**	**0.0228**	**0.3659**	**0.8816**

Note: 'D-' indicates that the model utilized the data preprocessing module.

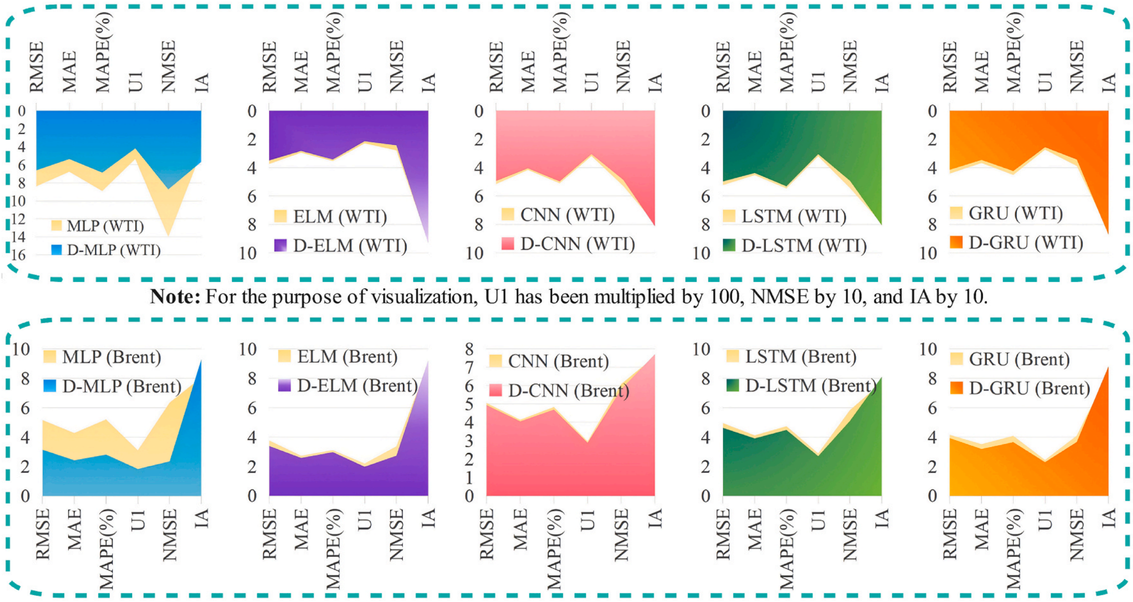

Fig. 3. Comparison of forecasting performance without/with data preprocessing.

Table 6
Prediction results of the proposed system and multiple models.

Dataset	Model	Model performance					
		RMSE	MAE	MAPE(%)	U1	NMSE	IA
WTI	**D-MLP**	6.6252	5.3637	6.8753	0.0418	0.8740	0.5680
	D-ELM	3.4929	2.8237	3.4270	0.0214	0.2429	0.9315
	D-CNN	4.9389	4.0499	4.9728	0.0305	0.4857	0.8133
	D-LSTM	4.9738	4.3808	5.3153	0.0305	0.4926	0.8063
	D-GRU	4.1490	3.4545	4.2388	0.0256	0.3428	0.8730
	DMEAI*	**3.0900**	**2.4903**	**3.0616**	**0.0191**	**0.1901**	**0.9409**
Brent	**D-MLP**	3.1374	2.4217	2.8105	0.0183	0.2347	0.9283
	D-ELM	3.3782	2.5688	2.9629	0.0197	0.2721	0.9219
	D-CNN	4.9379	4.0540	4.6958	0.0289	0.5813	0.7712
	D-LSTM	4.6268	3.9078	4.4847	0.0270	0.5103	0.8095
	D-GRU	3.9178	3.1792	3.6452	0.0228	0.3659	0.8816
	DMEAI*	**3.0831**	**2.3912**	**2.7710**	**0.0181**	**0.2266**	**0.9293**

Note: 'D-' indicates that the model utilized the data preprocessing module. DMEAI* indicates that the error-correction module is not applied.

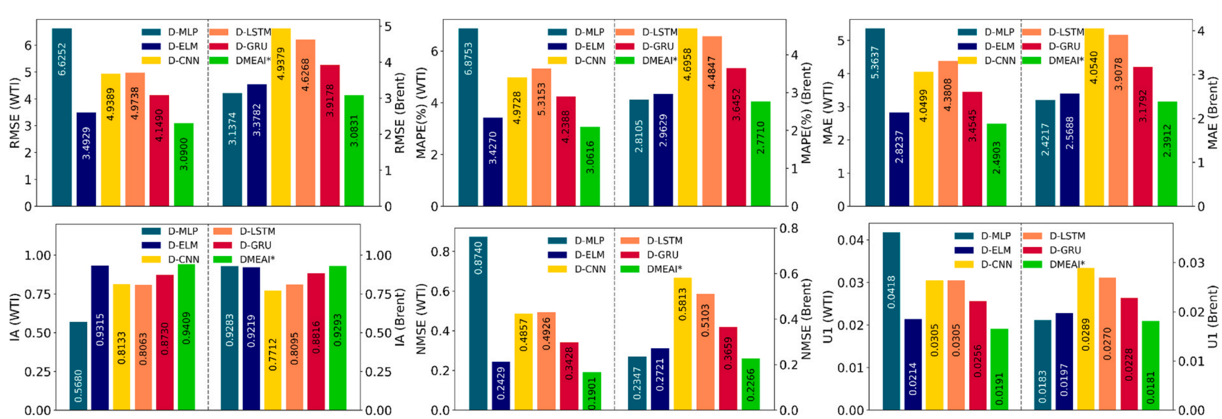

Fig. 4. Comparison of forecasting performance between the proposed system and multiple sub-models.

Brent, the DMEAI* system's RMSE is 3.0831, MAE is 2.3912, MAPE is 2.7710, U1 is 0.0181, NMSE is 0.2266, and IA is 0.9293. Compared with the best sub-model, these results indicated improvements of 1.7307 %, 1.2594 %, 1.4054 %, 1.0929 %, 3.4512 %, and 0.1077 %, respectively. Compared with those of the worst sub-model, the performance gains were 37.5625 %, 41.0163 %, 40.9898 %, 37.3702 %, 61.0184 %, and 20.5005 %, respectively.

Remark: Although the data preprocessing module enhances the predictive performance of the sub-models, one specific sub-model based on data preprocessing remains constrained by the inherent limitations of the models. Fortunately, the ensemble system developed in this study effectively combines the strengths of each sub-model in handling various data characteristics, achieving better predictive performance than any individual sub-model.

3.5. Experiment III: superiority of the designed ensemble modeling module over other advanced ensemble methods

In **Experiment II**, the proposed DMEAI* system demonstrated its superiority across various sub-models; however, its competitive edge with other advanced ensemble methods requires further validation. To achieve this, we subjected the DMEAI* to a comparative analysis with various ensemble models. Twelve distinct models were selected for comparison, representing the four major categories of mainstream ensemble methods: regression-based (Regression), simple aggregation (Simple), optimization-based (Optimizer), and stacking-based (Stacking) methods. Regression-based methods include the least absolute shrinkage and selection operator (LASSO) and ridge regression. The simple aggregation methods consisted of equal-weighted average (EWA); MEDIAN, which aggregates the predictions of the five sub-models using the median; and TRIM, which calculates the average after excluding the highest and lowest predictions. Optimization-based methods include the gray wolf optimizer (GWO) and multi objective gray wolf optimizer (MOGWO). Both optimization algorithms were used to determine the weights of the five sub-models. Stacking-based ensemble methods include CNN, long short-term memory (LSTM), gradient boosting regression (GBR), transformer, and neural basis expansion analysis for interpretable time series forecasting (N-BEATS)), which are advanced deep learning models designed specifically for time series prediction. All stacking-based models treated the five sub-models as base learners, with their respective architectures serving as meta-learners to produce the final ensemble forecasting results. For convenience, the twelve comparison models based on these ensemble methods are referred to as E-LASSO, E-Ridge, E-EWA, E-MEDIAN, E-TRIM, E-GWO, E-MOGWO, E-CNN, E-LSTM, E-GBR, E-Transformer and E-N-BEATS, where "E" denotes "ensemble." To ensure the fairness and comparability of the experimental outcomes, the proposed system without adaptive error correction, denoted as DEAMI*, was compared with nine ensemble methods. Fig. 5 presents a comparison of the true and predicted values for the different methods. The detailed experimental results are presented in Table 7 and Fig. 6 presents.

(a) The DMEAI* system outperformed the other ensemble models across various performance metrics, exhibiting the highest accuracy, lowest error, and greatest ability to capture trend changes. Some ensemble approaches even underperformed compared with sub-models. The Stacking and Simple types performed the worst on both datasets. The Optimizer and Regression types performed better but still performed worse than the proposed system.
(b) In the Stacking class, when E-LSTM was used as an example, the DMEAI* system achieved a 35.3205 % improvement in the RMSE for WTI, 27.75395 % for Brent, and a 39.4031 % enhancement in the MAPE for WTI and 33.9105 % for Brent. When benchmarked against state-of-the-art model-based ensemble methods, taking Brent as an example, the DMEAI* system still demonstrated superior forecasting accuracy, achieving an RMSE of 3.0900, which is approximately 48.0505 % lower than that of the E-transformer model and 48.8927 % lower than that of the E-N-BEATS. Similar advantages for DMEAI* were observed across other key metrics. These results indicate that Stacking models are more complex, with longer training times, higher resource consumption, and lower prediction accuracy, whereas the DMEAI* system maintains high precision while reducing complexity.
(c) In the Simple class, when E-TRIM was used as a case study, the DMEAI* system achieved enhancements of 29.2696 % in WTI and 18.9106 % in Brent, along with improvements of 31.6943 % in WTI and 23.8694 % in Brent. The performance of a simple class devoid of appropriate weighting is significantly inferior to the exceptional performance of the DMEAI* system.
(d) Compared with the optimizer class, the DMEAI* system exhibited a marked improvement in the RMSE, a pivotal metric, across both the WTI and Brent datasets. DMEAI* reduced the RMSE by approximately 6.0 % and 5.0 % for the WTI and Brent datasets, respectively. This enhancement indicates that DMEAI* is more effective at reducing prediction bias, thereby improving accuracy. Additionally, DMEAI* excels in the IA metric, which reflects model consistency. For the WTI dataset, the IA value of DMEAI*

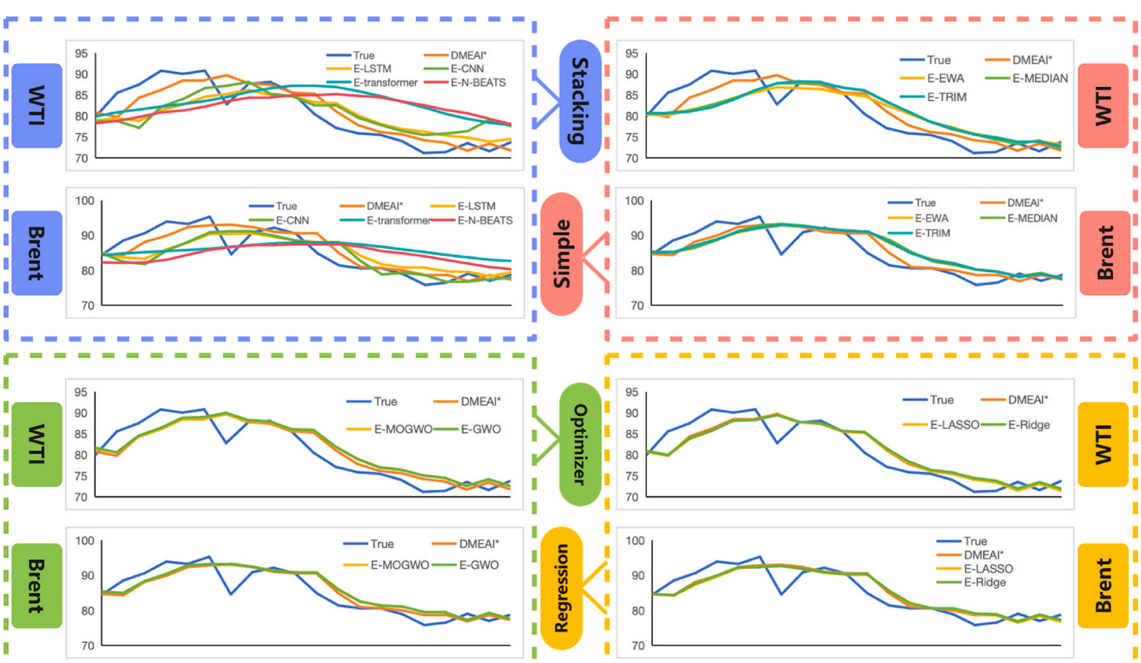

Fig. 5. Comparison chart of true and predicted values of the different ensemble methods.

Table 7
Comparison results with those of other advanced ensemble models.

Dataset	Ensemble type	Model	Model performance					
			RMSE	MAE	MAPE(%)	U1	NMSE	IA
WTI	Regression	E-LASSO	3.1605	2.5673	3.1599	0.0195	0.1989	0.9380
		E-Ridge	3.1993	2.6383	3.2406	0.0197	0.2038	0.9349
	Simple	E-EWA	4.1459	3.5188	4.3323	0.0256	0.3423	0.8579
		E-MEDIAN	4.2654	3.6018	4.4202	0.0263	0.3623	0.8621
		E-TRIM	4.3687	3.6521	4.4822	0.0269	0.3800	0.8526
	Optimizer	E-GWO	3.2905	2.7291	3.3400	0.0202	0.2156	0.9312
		E-MOGWO	3.2964	2.7580	3.3775	0.0203	0.2164	0.9292
	Stacking	E-LSTM	4.7774	4.0626	5.0524	0.0296	0.4309	0.8029
		E-CNN	5.0288	4.3613	5.4397	0.0310	0.4774	0.7831
		E-GBR	4.0836	3.4099	4.2093	0.0252	0.3320	0.8735
		E-transformer	6.9141	6.1077	7.3598	0.0421	0.9519	0.5038
		E-N-BEATS	7.3142	6.5796	8.0178	0.0448	1.0652	0.3374
	-	DMEAI*	**3.0900**	**2.4903**	**3.0616**	**0.0191**	**0.1901**	**0.9409**
Brent	Regression	E-LASSO	3.1670	2.5087	2.9149	0.0186	0.2391	0.9255
		E-Ridge	3.1472	2.4715	2.8674	0.0184	0.2361	0.9259
	Simple	E-EWA	3.7994	3.1797	3.6656	0.0222	0.3441	0.8827
		E-MEDIAN	3.9077	3.1975	3.6793	0.0228	0.3640	0.8797
		E-TRIM	3.8021	3.1590	3.6398	0.0222	0.3446	0.8842
	Optimizer	E-GWO	3.2411	2.5617	2.9660	0.0189	0.2504	0.9224
		E-MOGWO	3.2637	2.5886	2.9958	0.0191	0.2539	0.9212
	Stacking	E-LSTM	4.2675	3.5865	4.1928	0.0251	0.4114	0.8290
		E-CNN	4.3203	3.3745	3.9624	0.0255	0.4217	0.8572
		E-GBR	4.1738	3.1604	3.5857	0.0243	0.4153	0.8831
		E-transformer	5.9348	5.4201	6.3205	0.0347	0.8397	0.4042
		E-N-BEATS	6.0326	5.4223	6.4212	0.0355	0.8676	0.4620
	–	DMEAI*	**3.0831**	**2.3912**	**2.7710**	**0.0181**	**0.2266**	**0.9293**

Note: DMEAI* denotes that the error correction module was not applied.

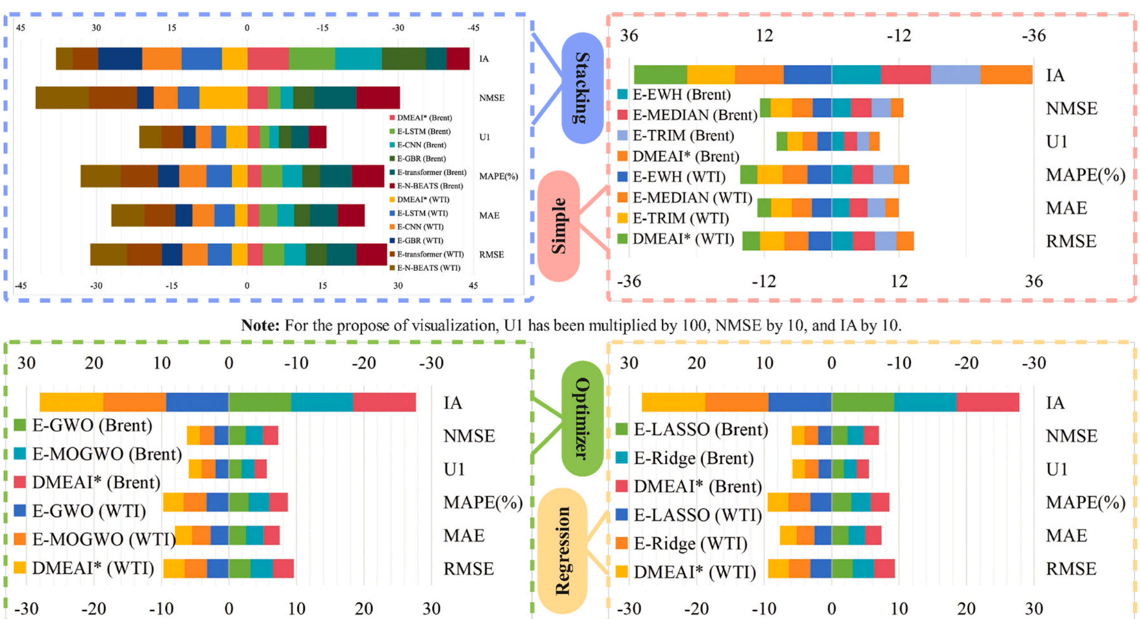

Fig. 6. Comparison of the forecasting performances of the proposed system and other ensemble methods.

surpassed that of the optimizer class by approximately 1.0 %–1.25 %. This improvement suggests that DMEAI* is superior in capturing long-term trends and avoiding predictions with significant deviations.

(e) Compared with the Regression class, the proposed system demonstrated the best performance and strongest robustness. For the WTI dataset, compared with E-LASSO and E-Ridge, DMEAI* reduced the RMSE by 2.2307 % and 3.4164 %, respectively. For the Brent dataset, compared with E-LASSO and E-Ridge, DMEAI* reduced the RMSE by 2.6492 % and 2.0367 %, respectively. Although the performance of the E-LASSO and E-Ridge models was close to that of DMEAI* on some metrics, their robustness across different datasets was inferior. E-LASSO slightly outperformed E-Ridge on the WTI dataset but underperformed on the Brent dataset. This fluctuation in performance indicates that the two models are sensitive to variations in data characteristics and struggle to adapt to complex fluctuations in crude oil prices under different market conditions. Consequently, their performances lack stability across various datasets, and the reliability and consistency of their predictions require improvement. By contrast, DMEAI* demonstrated greater robustness. On both the WTI and Brent datasets, DMEAI* not only achieves higher prediction accuracy across multiple metrics but also shows minimal performance variability, maintaining superior performance in different market environments. This suggests that DMEAI* can accurately predict price trends across different datasets and effectively address the variations among them, thereby demonstrating stronger robustness and adaptability.

Remark: Each ensemble method has strengths and limitations that affect the overall forecasting results. The Stacking class is complex with many parameters and remains constrained by the underlying machine learning models used. The Simple class, with an overly simplistic weight setting, failed to capture complex data patterns. The Optimizer class, which is capable of optimizing weight allocation, is prone to local optima, which limits its overall performance. In the Regression class, E-ridge helps smooth overfitting, whereas E-LASSO may overlook important features, thus limiting prediction accuracy and robustness. By contrast, DMEAI* can uncover hidden features and complex relationships across multiple sub-models, balancing different feature influences while fully utilizing global information. This enables DMEAI* to deliver superior predictive performance compared with other ensemble methods.

3.6. *Experiment IV: effectiveness of the designed adaptive error correction module*

This section begins with an analysis of the results of applying the error correction decision to the WTI and Brent datasets and then analyzes the effectiveness of the error compensation mechanism for datasets requiring error correction. The detailed outcomes of the cross-validation are shown in Fig. 7. In the case of WTI, $AMSE_0$ is less than $AMSE_1$, indicating that the error compensation mechanism should not be applied to the test set data for WTI. Hence, the predictions from the ensemble process serve as the final predictions for the forecasting system. Conversely, for Brent, $AMSE_0$ exceeds $AMSE_1$, suggesting that an error-compensation mechanism is warranted for the Brent test set data. Table 8 presents a comparative analysis of the outcomes before and after applying the error-compensation mechanism. Before the error compensation mechanism, denoted as DMEAI* was applied, the RMSE, MAE, and MAPE values were 3.0831, 2.3912, and 2.7710 %, respectively. After applying this method, denoted as DMEAI, the RMSE, MAE, and MAPE were 3.0719, 2.3172, and 2.6774 %, respectively, indicating decreases of 0.3633 %, 3.0947 %, and 3.3778 %, respectively. Similarly, after the error compensation mechanism was applied, the values of U1

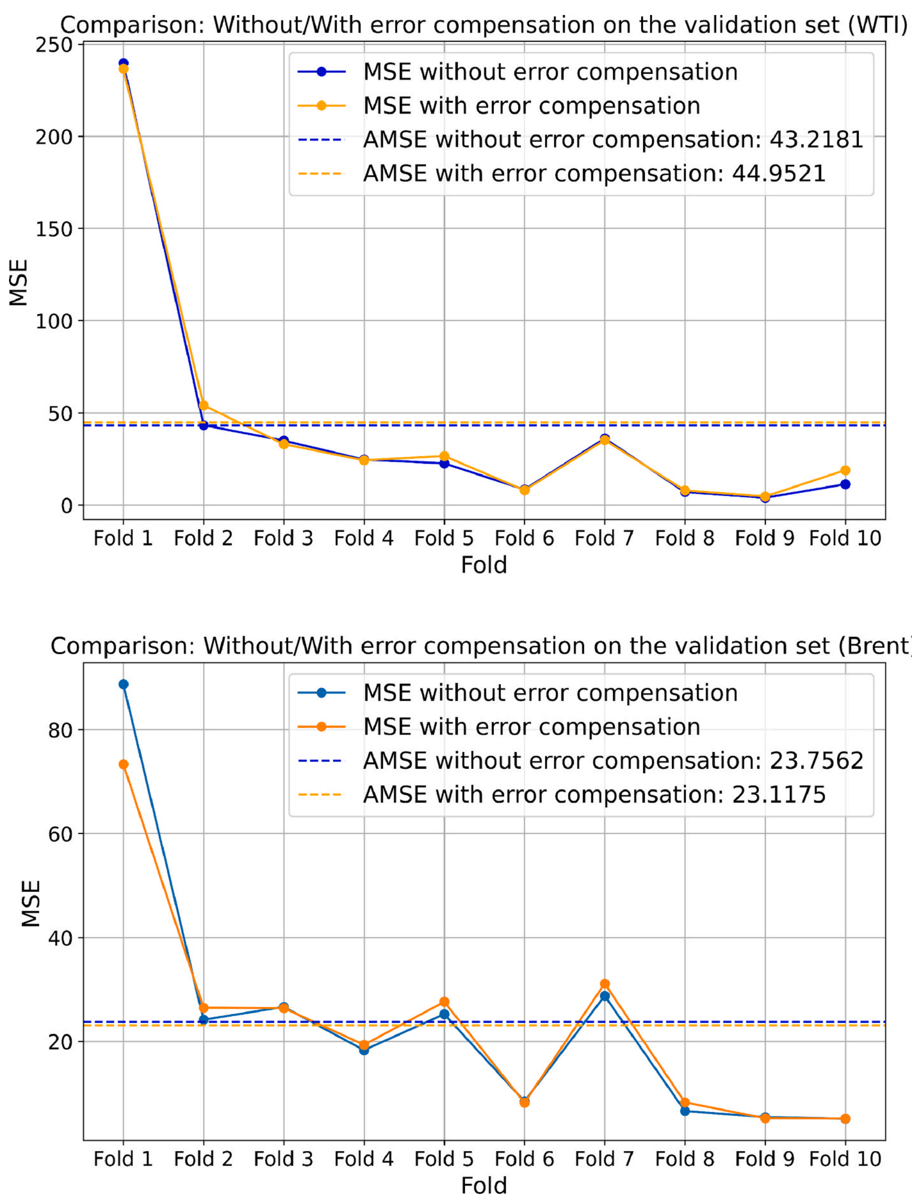

Fig. 7. Comparison of cross-validation results on the validation set.

Table 8
Prediction results of the developed ensemble system before and after error compensation.

Model	Model performance					
	RMSE	MAE	MAPE(%)	U1	NMSE	IA
DMEAI*	3.0831	2.3912	2.7710	0.0181	0.2266	0.9293
DMEAI	**3.0719**	**2.3172**	**2.6774**	**0.0180**	**0.2250**	**0.9317**

and NMSE decreased by 0.5525 % and 0.7061 %, respectively, whereas the value of IA increased by 0.2583 %. The error compensation mechanism led to improvements in the overall performance, thereby verifying its effectiveness.

Remark: These results confirm that the designed adaptive error-correction module effectively improves the forecasting accuracy for datasets requiring error correction. Notable reductions in the RMSE, MAE, MAPE, U1, and NMSE, along with consistent performance across metrics, demonstrate its robustness and reliability. Datasets that do not require correction reflect the concept of the proposed adaptive error compensation strategy.

3.7. *Experiment V: ablation study*

Ablation experiments were conducted to systematically assess the independent contributions and necessity of each key module in the DMEAI system for its overall predictive performance. In each comparison, one core module was either removed or replaced, whereas all other components were kept intact or set to their optimal configurations, allowing for a clear quantification of the effect of the ablated module. Model 1 excludes the outlier detection and correction components but retains the normalization and denormalization processes. After training and prediction via the five sub-models, the proposed ensemble method and adaptive error correction module were applied. In Model 2, the ensemble modeling module was removed, replacing with the best-performing sub-model and applying the adaptive error correction module. Models 3-4 adopted other ensemble methods combined with an adaptive error-correction module. Specifically, these methods include the ensemble method, which ranked second to the proposed method in **Experiment III**. Model 5 excluded the adaptive error-correction module, which corresponded to the results presented in **Experiment IV**. The results of the ablation study presented in Table 9 clearly demonstrate the integral role and significant contribution of each core module to the overall performance of the proposed DMEAI system. A detailed analysis follows.

(a) When the data preprocessing module was removed (Model 1), a noticeable degradation in forecasting accuracy was observed across both the WTI and Brent datasets, highlighting the importance of outlier detection and correction for more accurate predictions. For example, on the WTI dataset, the DMEAI reduced the RMSE from 3.4018 to 3.0900, resulting in a performance improvement of 9.1657 %, and lowered the MAE from 3.1074 to 2.4903, achieving an improvement of 19.8590 %. Similarly, for the Brent dataset, the RMSE was reduced by 14.1280 %, and the MAE decreased by 16.6415 %, highlighting the substantial gains brought by the outlier correction module.

(b) Similarly, altering the ensemble modeling module using the best-performing individual sub-model or other strong alternative ensemble methods, consistently resulted in suboptimal performance relative to the integrated DMEAI system. Compared with Model 2, which removed the ensemble modeling module and relies solely on the best-performing sub-model combined with the error correction mechanism, DMEAI achieved consistently better results. For example, for the WTI dataset, the RMSE and MAE decreased by 3.5189 % and 2.5285 %, respectively. For the Brent dataset, the improvements were similar, with a 2.0877 % reduction in the RMSE and a 4.3152 % reduction in the MAE. This underscores the effectiveness of our ensemble method in synergizing the diverse strengths of the sub-models.

(c) Furthermore, an evaluation of the system without the adaptive error correction module (Model 5, i.e., DMEAI*) confirmed its value. For the Brent dataset, in which error correction was deemed necessary, a discernible decline in accuracy was evident when this module was not applied, reaffirming the benefits of the adaptive error compensation strategy. It should be clarified that although DEMAI and Model 5 yield the same prediction results on the WTI dataset, this does not signify the ineffectiveness of the adaptive correction module. Rather, the fact that this dataset's predictions inherently require no error correction serves as a validation of the module's designed effectiveness.

Remark: In summary, these ablation experiments confirm that the complete DMEAI system, with all its modules functioning cohesively, achieves the best forecasting performance, validating the design choices and necessity of each key component in the proposed framework.

4. Discussion

This section provides a broader discussion of the characteristics and implications of the system. Section 4.1 presents an in-depth interpretable analysis. Section 4.2 extends the empirical evaluation to a different market context to assess the system's performance, generalizability for Asian markets, and capabilities in long-term forecasting. Section 4.3 explores the practical operationalization of interpretability insights. Finally, Section 4.4 discusses the time cost of the proposed system.

Table 9
Prediction results of the ablation study.

Dataset	Model	Model performance					
		RMSE	MAE	MAPE(%)	U1	NMSE	IA
WTI	**Model 1**	3.4018	3.1074	3.8154	0.0209	0.2304	0.9244
	Model 2	3.2027	2.5549	3.1311	0.0197	0.2042	0.9401
	Model 3	3.1993	2.6383	3.2406	0.0197	0.2038	0.9349
	Model 4	3.1605	2.5673	3.1599	0.0195	0.1989	0.9380
	Model 5	**3.0900**	**2.4903**	**3.0616**	**0.0191**	**0.1901**	**0.9409**
	DMEAI	**3.0900**	**2.4903**	**3.0616**	**0.0191**	**0.1901**	**0.9409**
Brent	**Model 1**	3.5773	2.7798	3.1987	0.0210	0.3051	0.9007
	Model 2	3.1374	2.4217	2.8105	0.0183	0.2347	0.9283
	Model 3	3.1472	2.4715	2.8674	0.0184	0.2361	0.9259
	Model 4	3.1670	2.5087	2.9149	0.0186	0.2391	0.9255
	Model 5	3.0831	2.3912	2.7710	0.0181	0.2266	0.9293
	DMEAI	**3.0719**	**2.3172**	**2.6774**	**0.0180**	**0.2250**	**0.9317**

4.1. Interpretable analysis

To enhance the interpretability of the predicted results, an interpretable analysis module was employed to perform global and local interpretability analyses.

4.1.1. Global interpretability analysis

Figs. 8. (a) and (b) show the impact of each sub-model on the prediction values, where red denotes a higher prediction value by the sub-model and blue indicates a lower prediction value. The color represents the magnitude of the sub-model's prediction, which is unrelated to the SHAP values. The horizontal axis represents the magnitude of the SHAP values, which quantify the impact of each sub-model on the output of the proposed ensemble system. The vertical axis lists the names of all sub-models arranged in descending order of importance on the basis of the absolute value of their SHAP values, with those at the top having larger distributions and greater importance.

(a) For the WTI dataset, the descending order of feature importance was D-MLP > D-ELM > D-GRU > D-CNN > D-LSTM. D-MLP had the most significant impact on the ensemble system, whereas D-CNN and D-LSTM contributed less to the overall system. When the prediction of D-MLP for a particular sample is high, indicated by a red point, the corresponding SHAP values are mostly positive, suggesting a positive contribution to the ensemble system. Conversely, when the prediction of D-MLP was low, as indicated by the blue point, the SHAP values were mostly negative, indicating a negative contribution to the ensemble system. The behavior of D-CNN is opposite to that of D-MLP.

(b) For the Brent dataset, D-MLP continued to exert the most significant influence on the ensemble system, whereas D-ELM, which contributed significantly to the WTI predictions, had the lowest contribution. For example, when the prediction for a particular sample is low, as indicated by the blue point, the SHAP values are mostly positive, indicating a positive contribution of D-ELM to the ensemble system. Conversely, when the prediction for a sample was high, as indicated by the red points, the SHAP values were mostly negative, indicating a negative contribution of D-ELM to the ensemble system.

(c) Compared with the global interpretability plot, the heatmap plot provides a more detailed representation of the distribution of SHAP values across individual samples. Consequently, the heatmap plot constitutes a matrix of SHAP values, as illustrated in Figs. 8. (c) and (d). The horizontal axis of the heatmap denotes each sample, whereas the ordinate represents distinct sub-models. Each row shows the SHAP value distribution for a particular sub-model. The color bar to the right of the plot shows the range of SHAP values, and the black functional curve at the top of the graph represents the predictive values of the ensemble model.

4.1.2. Local interpretability analysis

Summary and heatmap plots were primarily utilized for global analysis to shed light on the overall impact of all samples or sub-models on the predictive outcomes of the ensemble system. To provide a local interpretability analysis, a waterfall plot, presented in Fig. 9, was used to delineate the influence of each sub-model on the predictive outcome of an individual sample in a stepwise manner.

For the WTI and Brent datasets, a representative sample was selected for detailed local interpretability analysis. At the base of the waterfall plot, *E[f(x)]* represents the baseline prediction value, indicating the predictive value of the ensemble system in the absence of sub-models. The red blocks indicate that the respective sub-models increased the predictive value of the ensemble system, whereas the blue blocks indicate a decrease in the predictive value of that sub-model. Concurrently, the length of each block corresponds to the absolute magnitude of the SHAP value of the sub-model. Taking the WTI dataset as an example, *E[f(x)]* is 83.962, which implies that without any sub-models, the predictive value of the system would be 83.962. D-CNN augmented the ensemble system prediction by 0.12, and D-LSTM, D-GRU, D-ELM, and D-MLP decreased the ensemble system prediction by 0.32, 0.47, 1.48, and 2.09, respectively. The cumulative progression of the SHAP values from the five sub-models resulted in an ensemble forecasting value of 79.729, as indicated by the upper portion of the waterfall plot. The waterfall plot of this sample reveals that there are significant differences in the contributions of the different sub-models to the ensemble system prediction. This visual local interpretation helps to better understand the roles of each sub-model in the ensemble forecasting system and their relative contributions to the final result.

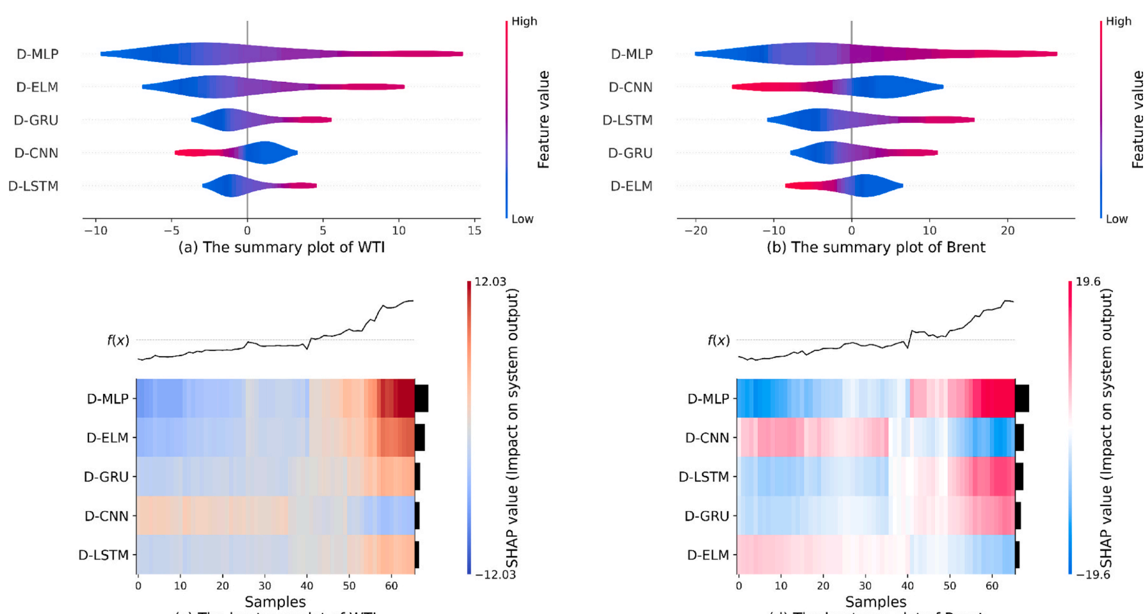

Fig. 8. Global interpretability analysis plot for the proposed system.

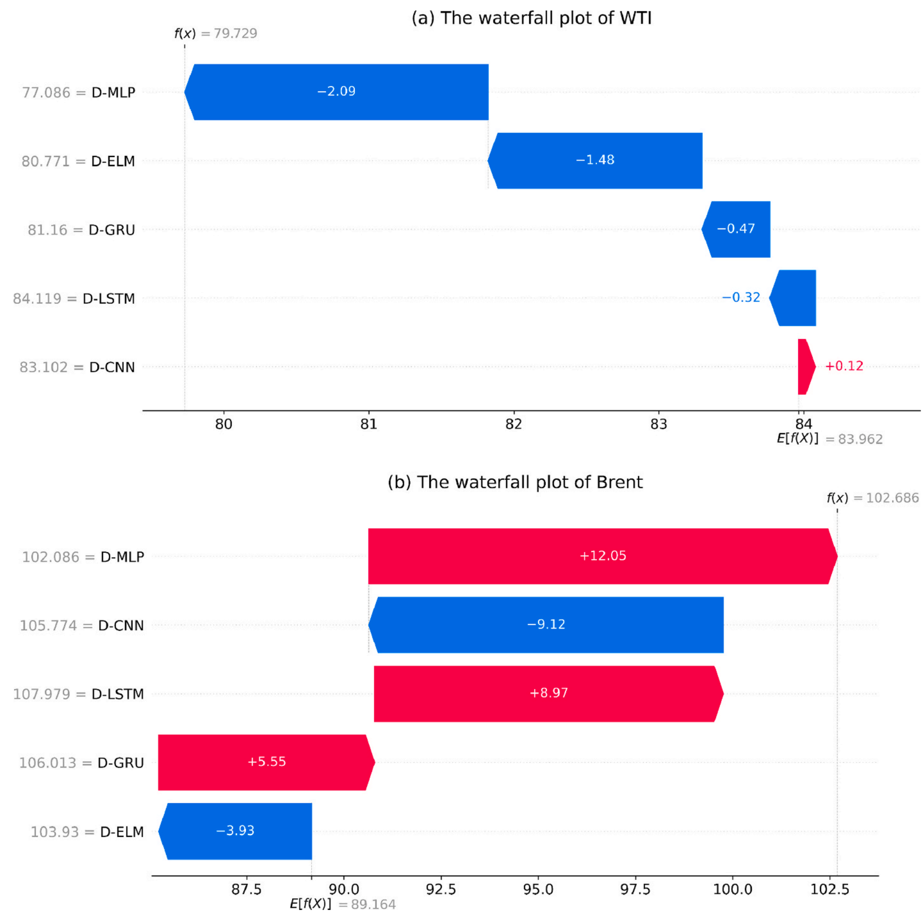

Fig. 9. Local interpretability analysis plot for the proposed system.

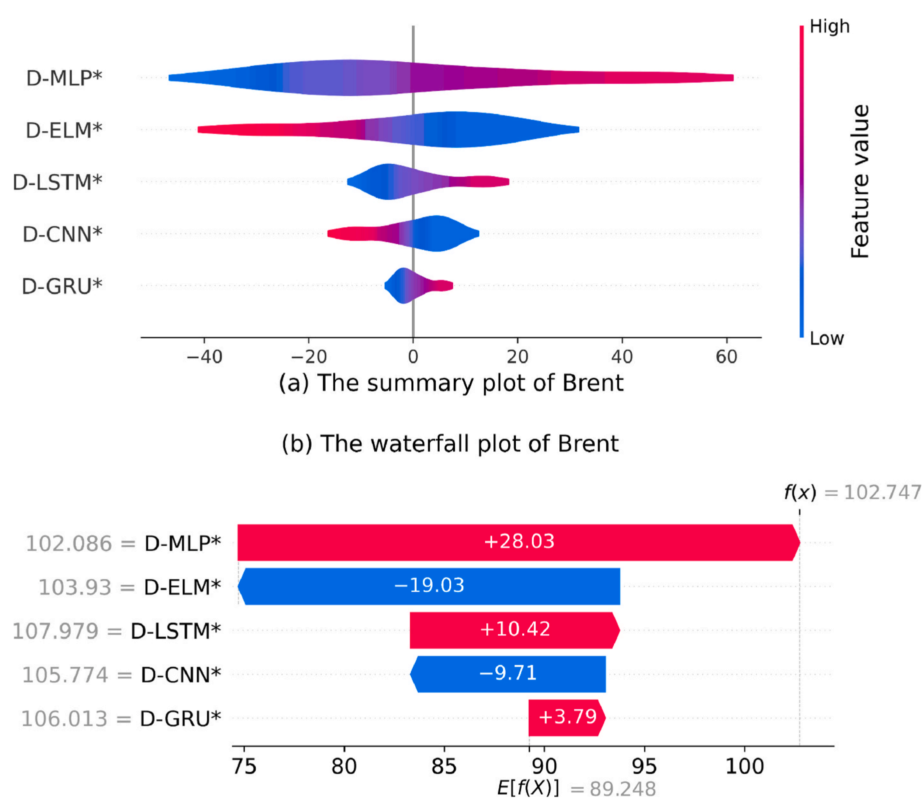

Fig. 10. Interpretability analysis plot for the proposed system applying adaptive error correction module on Brent.

4.1.3. Interpretable analysis following error compensation mechanism

Given the change in the error compensation mechanism of the original ensemble system, this section further examines the interpretability of the final system. As depicted in Fig. 10, the final system exhibits significant shifts in the importance of its constituent sub-models. To distinguish the systems that did not apply the adaptive error correction module, all the sub-models mentioned in this section are marked with an asterisk (*). Notably, D-ELM*, which was initially ranked lowest in importance, now ranks second. Concurrently, there was a marked alteration in the distribution of SHAP values across the sub-models. Compared with those of the original system, the SHAP values for D-ELM* in the final system were more dispersed, whereas those for D-CNN*, D-LSTM*, and D-GRU* were more concentrated. This suggests that the effect of D-ELM* on the final system is relatively unpredictable, whereas the effects of D-CNN*, D-LSTM*, and D-GRU* are more consistent. Fig. 10. (b) shows a sample identical to that shown in Fig. 9. (b), which illustrates that *E[f(x)]* shifted from 89.164 to 89.248, and the contribution of each sub-model to the final predicted value was also altered. These shifts were the direct result of the applied error-compensation mechanism. In summary, the error compensation mechanism accounts for potential errors in the original ensemble system, effectively rebalancing the contributions of different sub-models and thereby enhancing the predictive performance.

4.2. Evaluating model performance and generalizability: insights from Dubai crude oil

This section aims to validate the performance and generalizability of the proposed DMEAI system in the context of long-term forecasting and its applicability to the Asian crude oil market, as represented by Dubai crude oil. For this purpose, weekly Dubai crude oil price data spanning April 27, 2014, to December 29, 2024, comprising 558 data points, were utilized. This dataset was partitioned into training, validation, and test sets at a 7:2:1 ratio, resulting in a test set of 56 data points for performance evaluation. In this validation, the long-term forecasting performance of the DMEAI system was compared with that of its best-performing sub-model and 12 ensemble methods. The forecasting performances of the DMEAI system and selected benchmark models on the Dubai crude oil dataset, particularly those focused on long-term predictive horizons, are presented in Table 10.

(a) The proposed DMEAI system consistently demonstrated the highest forecasting accuracy across most evaluation metrics. It achieved the lowest RMSE (2.2838), MAE (1.5034), MAPE (1.8709 %), U1 (0.0144), and NMSE (0.1951). This indicates a strong ability to capture long-term trends in the Dubai crude oil price, which serves as a crucial indicator for the Asian market. The high IA value of 0.9473 further underscores the excellent agreement between the DMEAI predictions and actual price movements.

(b) Compared with the best-performing individual sub-models, the DMEAI system showed a substantial improvement, with an approximately 23.8479 % reduction in the RMSE and a 36.8067 % reduction in the MAPE. This significant margin highlights the effectiveness of our data preprocessing, ensemble modeling, and adaptive error compensation strategies in the context of long-term forecasting for this oil market.

(c) The DMEAI system outperformed state-of-the-art model based ensemble models, including E-transformer and E-N-BEATS. For example, DMEAI exhibited an RMSE approximately 73.4541 % lower than that of E-transformer and 39.4844 % lower than that of E-N-BEATS. Furthermore, compared with the E-ridge and E-LASSO ensemble methods, which were strong performers in earlier experiments, the DMEAI system maintained a competitive edge. Specifically, DMEAI achieved an MAE of 1.5034, representing improvements of approximately 6.0199 % and 5.3513 % over E-ridge and E-LASSO, respectively. Similar improvements were observed for the other metrics. This further validates the superior performance of our DMEAI system in long-term forecasting and in the Asian market, especially compared to other ensemble methods.

4.3. Practical implications of interpretability

Beyond quantitative forecasting accuracy, understanding why a model makes certain predictions is crucial to building trust and enabling informed action. This section discusses the practical applications of interpretability analysis in the DMEAI system. We illustrate how these insights can be operationalized by policymakers, investors, and other stakeholders in the crude oil market via several scenarios on the basis of our empirical SHAP findings.

Our interpretable analysis identified an important finding: the individual predictive accuracy of a sub-model does not necessarily correspond to its contribution to the final ensemble forecast. This finding provides critical insights for decision-makers, cautioning that excessive reliance on a single sub-model with superior individual performance may be misleading. The strength of an ensemble lies in its ability to harness diverse perspectives. A sub-model that appears less accurate in isolation might capture unique market dynamics or signaling components that other more accurate models miss. For an investor, focusing solely on one type of analytical model may lead to missing less frequent but crucial market signals that a simpler or different type of model within the ensemble contributes. For risk managers, understanding that different models are influential underpins the need for a diversified

Table 10
Prediction results for Dubai crude oil.

Ensemble type	Model	Model performance					
		RMSE	MAE	MAPE(%)	U1	NMSE	IA
-	**DMEAI**	**2.2838**	**1.5034**	**1.8709**	**0.0144**	**0.1951**	**0.9473**
-	**Best sub-model**	2.9990	2.3376	2.9606	0.0191	0.3365	0.8819
Regression	**E-Ridge**	2.3427	1.5997	1.9902	0.0147	0.2053	0.9453
	E-LASSO	2.3385	1.5884	1.9754	0.0147	0.2046	0.9454
Simple	**E-EWA**	3.2615	2.6675	3.4147	0.0208	0.3979	0.8661
	E-MEDIAN	3.4247	2.7613	3.5359	0.0219	0.4388	0.8498
	E-TRIM	3.3447	2.7202	3.4813	0.0213	0.4185	0.8582
Optimizer	**E-GWO**	3.4921	2.9381	3.7909	0.0223	0.4562	0.0681
	E-MOGWO	3.2480	2.6617	3.4063	0.0207	0.3947	0.8716
Stacking	**E-LSTM**	4.2915	3.6012	4.4181	0.0265	0.6890	0.7572
	E-CNN	2.8460	2.3165	2.9487	0.0181	0.3030	0.8997
	E-GBR	3.2316	2.5484	3.2059	0.0203	0.3907	0.8509
	E-transformer	8.6032	7.4357	8.6308	0.0520	2.7689	0.4498
	E-N-BEATS	3.7739	3.1093	3.8102	0.0234	0.5328	0.8419

approach to risk assessment rather than trusting a single indicator.

Furthermore, this highlights the true value of the ensemble, beyond mere accuracy aggregation. The SHAP values demonstrate how the ensemble intelligently leverages the correct information from each sub-model at potentially different times or under different conditions. For policymakers, this implies that complex market phenomena such as crude oil prices are better understood and predicted by synthesizing information from multiple analytical viewpoints represented by diverse sub-models, rather than relying on a monolithic approach. This underscores the “no free lunch” theorem in practice that different models excel at different aspects, and a well-constructed ensemble, as interpreted by SHAP, shows how these varying strengths are optimally combined.

Another important finding from the interpretable analysis is the dynamic nature of sub-model contributions: although a sub-model may be more important in the global interpretable analysis, another sub-model that may not be as important globally becomes the main driver of the prediction in the local interpretable analysis of a specific sample point. This is highly actionable for investors and traders. For example, if the globally dominant sub-model suddenly shows a low local SHAP value for an upcoming critical forecast point, whereas a typically less influential model shows a high local SHAP value, it could indicate that the market is currently driven by atypical factors. This insight could prompt investors to reevaluate their confidence in the forecast, seek corroborating information, or adjust their hedging strategies.

4.4. Computational cost

The computational cost is an important factor in real applications. On the one hand, the proposed forecasting system has acceptable requirements for the operating environment. In this study, all the experiments were performed using a computing system featuring a 12th Gen Intel(R) Core (TM) i5-1240P processor (1.70 GHz) and integrated Intel (R) Iris(R) Xe Graphics. The primary software environment was Python 3.11. On the other hand, the runtime of the proposed forecasting system is within an acceptable range. Specifically, the data preprocessing stage was highly efficient, requiring approximately 0.04 s for each dataset. Similarly, the subsequent ensemble modeling stage and adaptive error correction stages were computationally inexpensive, with their combined training and prediction processes completed in approximately 0.07 s. For the multiple-model forecasting stage, each of the five sub-models was independently trained and used for five predictions. This repetition was implemented to account for potential stochasticity in the training processes and ensure the robustness of the performance evaluation, with the final sub-model predictions fed into the ensemble as the average of these five runs. In a single run, the average combined training and prediction time for the longest running sub-model was approximately 6 s. In summary, the time consumption of the entire DMEAI system was approximately 30 s, which is suitable for real-time applications.

5. Conclusions

Crude oil price forecasting is an important research topic in economics, finance, and energy. This study aims to address the uncertainty of price fluctuations in the crude oil market effectively and provide reliable and effective information for energy policy formulation, corporate strategic planning, and investment decision-making, thereby promoting the stability and sustainable development of the global energy market. This study proposes a crude oil price ensemble forecasting system based on outlier correction and an adaptive error-compensation strategy. It comprises five modules: data preprocessing, multiple-model forecasting, ensemble modeling, adaptive error correction, and interpretability analysis. The data preprocessing module was designed to identify and correct outliers in the original data, thereby eliminating their negative impact on model training and forecasting performance. The multiple model forecasting module was designed to provide diverse sub-models for crude oil forecasting. Unlike most previous ensemble forecasting studies, an effective ensemble modeling method was introduced to achieve better results than any individual sub-model by fully leveraging its strengths. Another key contribution of this study is the proposal of an adaptive error-compensation strategy, upon which an adaptive error-correction module is developed. The core idea of this strategy is to “test before correcting”—it first assesses whether error correction is necessary and then applies the correction if needed. The interpretable analysis module provides key insights and practical implications to decision makers by enhancing the transparency of the ensemble system. Empirical research on the WTI and Brent crude oil futures markets demonstrates that the developed system outperforms all the compared models, making it a promising solution for crude oil price forecasting.

The specific findings are as follows. (1) The ensemble forecasting system exhibits exceptional performance in crude oil price forecasting, exhibiting high accuracy and robustness, with RMSE values of 3.0900 and 3.0719 for WTI and Brent, respectively, and MAPE values of 3.0616 % and 2.6774 % for WTI and Brent, respectively. (2) The data preprocessing module significantly eliminated the negative impact on model training and forecasting performance. For WTI and Brent, the improvements in the RMSE across the different sub-models ranged from 4.1660 % to 21.0299 % and from 2.2972 % to 39.1788 %, respectively. (3) The proposed ensemble system outperformed other advanced ensemble methods in terms of accuracy, complexity, and robustness. This study compared twelve models, which were divided into four types: Regression, Simple, Optimizer, and Stacking. The proposed system achieved the best performance across all the evaluation metrics by fully using each sub-model’s strengths. The proposed system demonstrates excellent robustness and consistently performs well across different datasets. By contrast, the second-best ensemble method in terms of prediction accuracy exhibited poor robustness and inconsistent performance across various datasets. (4) An adaptive error compensation strategy and the corresponding error correction module developed in this study were designed to assess the necessity of error correction and enhance prediction performance when such corrections are warranted. The system demonstrated improved predictive accuracy and robustness when datasets were evaluated prior to application. (5) The interpretable analysis module simultaneously considers the scenarios before and after error correction to clarify the ensemble stage. It also provides decision-makers with a clear explanation of the ensemble process with applications to enhance their understanding of how different modeling approaches can be jointly used for prediction.

Although the forecasting system proposed in this study has demonstrated superior predictive performance for crude oil prices compared to benchmark models, several areas warrant further in-depth investigation to enrich research in this domain. Firstly, future research could expand the dimensionality of interpretability by incorporating techniques that trace predictions back to the underlying input features or influential data points. This would provide a more granular understanding of the decision sources driving both the individual base models and, consequently, the ensemble forecast. Secondly, exploring online learning and reinforcement learning methodologies presents a promising avenue. Such approaches could enable the system to adapt more dynamically to new market data and evolving conditions, potentially enhancing predictive responsiveness and accuracy in real-time applications.

CRediT authorship contribution statement

Wendong Yang: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Weicheng Ma:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Bo Zeng:** Writing – review & editing, Supervision, Methodology. **Yan Hao:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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